Logic answer key (version 1.0.0)

Private grading material. Each entry gives the question ID, its page in the question collection, the correct response, the full reasoning, why each wrong choice fails, accepted alternatives, the grading rule and how the key was verified.

S1-M-Q1 · question page 2

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

The shaded shape is reflected in the dashed mirror line on its right. Which figure shows its mirror image? (Only the shape matters, not where it sits in its box.)

Answer: C. Figure C

Reasoning: A reflection in a vertical line keeps every cell in its row and swaps left and right. Figure C is the only panel whose rows match the shape row by row with left and right swapped.

- ✗ Figure A: a half-turn of the shape, not a reflection
- ✗ Figure B: the reflection in a horizontal line (top and bottom swapped)
- ✗ Figure D: a quarter-turn of the shape

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S1-M-Q2 · question page 3

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

Each figure shows a shape above a dashed water line and, below the line, what is supposed to be its reflection in the water. In which figure can the lower shape NOT be the reflection of the upper one?

Answer: D. Figure D

Reasoning: A reflection in a horizontal line keeps every cell in its column and reverses the order of the rows, measured from the line. In D the lower

shape is the upper shape moved down without flipping, so the row nearest the line is wrong.

- ✗ Figure A: the lower shape is the upper shape flipped top to bottom across the line, so it is a real reflection
- ✗ Figure B: the lower shape is the upper shape flipped top to bottom across the line, so it is a real reflection
- ✗ Figure C: the lower shape is the upper shape flipped top to bottom across the line, so it is a real reflection

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S1-M-Q3 · question page 4

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

Words are written left to right in block capitals and seen in a left-right mirror (a vertical mirror line). In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same in the mirror?

Answer: A. TOOT

Reasoning: A left-right mirror reverses the order of the letters and mirrors each letter. The word looks the same only if it reads the same backward AND every letter is a listed symmetric letter. TOOT is the only choice that does both.

- ✗ NOON: N is not one of the listed letters, so each N appears reversed in the mirror
- ✗ ZOOM: Z is not listed, and the mirror reverses the letter order, giving MOOZ
- ✗ BIB: B is not listed, so each B appears reversed even though BIB reads the same backward

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the

check function accepts only the keyed choice.

S1-M-Q4 · question page 5

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

A word is written in block capitals and seen reflected in calm water below it (a horizontal mirror line). In the block capitals used here, exactly these letters look the same in a top-bottom mirror: B C D E H I K O X. The letter order stays the same in this reflection. Which word looks exactly the same in the water?

Answer: B. CODE

Reasoning: A top-bottom reflection keeps the order of the letters and flips each letter. The word looks the same when every letter is a listed letter; only CODE qualifies.

- ✗ CORE: R is not one of the listed letters, so it appears upside down
- ✗ BEAD: A is not listed, so it appears upside down
- ✗ BOXES: S is not listed, so it appears changed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S1-M-Q5 · question page 6

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

The grid has 4 rows and 8 columns. The dashed mirror line runs between columns 4 and 5. The shaded cells are in row 1 column 2, row 2 columns 2, 3 and 4, and row 3 column 4. Mark the cells of their mirror image. (Cells are named row,column; row 1 is the top row and column 1 the left column.)

Answer: 1,7, 2,5, 2,6, 2,7, 3,5

Reasoning: Each cell keeps its row, and a cell in column c lands in column 9 - c (the same distance on the other side of the line between columns 4 and 5): (1,7), (2,7), (2,6), (2,5), (3,5).

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: the key is computed as column 9 - c for every shaded cell.

S1-B1-Q1 · question page 7

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A grid has 10 columns. A vertical mirror line lies between columns 5 and 6. A shaded cell is in column 2. In which column is its mirror image?

Answer: 9

Reasoning: Column 2 is 4 columns left of the line (columns 2, 3, 4, 5), so its image is 4 columns right of it: column 9. Check: 2 + 9 = 11 = 5 + 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-B1-Q2 · question page 8

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A grid has 8 rows. A horizontal mirror line lies between rows 4 and 5. A shaded cell is in row 1. In which row is its mirror image?

Answer: 8

Reasoning: Row 1 is 4 rows above the line, so the image is 4 rows below it: row 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-B1-Q3 · question page 9

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A reflection in a vertical mirror line keeps every cell in its own row.

Answer: True

Reasoning: Only the column changes; the row stays the same.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S1-B1-Q4 · question page 10

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A cell is 3 columns to the left of a vertical mirror line. Where is its mirror image?

Answer: D. 3 columns to the right of the line

Reasoning: A reflection keeps the distance to the line and changes the side.

- ✗ 3 columns to the left of the line: that is the original cell
- ✗ 6 columns to the right of the line: the distance from the line is kept, not doubled
- ✗ on the mirror line itself: only points on the line stay on it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S1-B1-Q5 · question page 11

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A row of 6 cells has a mirror line between cells 3 and 4. Cell 1 is shaded. Mark the cell of its mirror image. (The grid is 1 row by 6 columns; cells are named row,column.)

Answer: 1,6

Reasoning: Cell 1 is 3 cells left of the line, so its image is 3 cells right: cell 6.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: image column = 7 - 1.

S1-F1-Q1 · question page 12

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A clock face with no numbers shows 2:20. You see it in a left-right mirror. What time does the mirrored clock appear to show?

Answer: C. 9:40

Reasoning: A left-right mirror sends a time t to 12:00 - t. 12:00 - 2:20 = 9:40. Check: the minute hand at 4 (20 minutes) appears at 8 (40 minutes), and the hour hand just past 2 appears just before 10.

- ✗ 10:20: only the hour hand was mirrored; the minute hand was left at 20 minutes
- ✗ 2:40: only the minute hand was mirrored
- ✗ 3:40: the hands were swapped instead of mirrored

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S1-F1-Q2 · question page 13

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

The shape is reflected in a vertical mirror line and then the result is reflected in a horizontal mirror line. Which figure shows the final result? (Only the shape matters, not where it sits.)

Answer: B. Figure B

Reasoning: A vertical reflection then a horizontal one sends (x, y) to (-x, -y): a half-turn. Figure B is the shape turned halfway around.

- ✗ Figure A: only the first (vertical) reflection
- ✗ Figure C: the starting shape: the two reflections do not cancel
- ✗ Figure D: only the second (horizontal) reflection

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S1-F1-Q3 · question page 14

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A word is written DOWN the page, one block capital under another, and seen in a left-right mirror. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same?

Answer: A. MATH

Reasoning: Written downward, a left-right mirror keeps the order (top to bottom) and mirrors each letter, so the word looks the same when every letter is listed. Only MATH qualifies.

- ✗ MAZE: Z and E are not listed, so they appear reversed
- ✗ BATH: B is not listed
- ✗ MASK: S and K are not listed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S1-F1-Q4 · question page 15

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A 6 by 6 grid has two dashed mirror lines: a vertical one between columns 3 and 4 and a horizontal one between rows 3 and 4. The figure must look the same in BOTH mirrors. Four cells are already shaded in the top-left quarter: (1,1), (1,2), (2,2) and (3,3). Mark every shaded cell of the completed figure, including the four given cells.

Answer: 1,1, 1,2, 1,5, 1,6, 2,2, 2,5, 3,3, 3,4, 4,3, 4,4, 5,2, 5,5, 6,1, 6,2, 6,5, 6,6

Reasoning: Each given cell (r, c) also needs its images (r, 7 - c), (7 - r, c) and (7 - r, 7 - c). That gives 4 x 4 = 16 cells, one copy of the pattern in

each quarter.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: the key is the union of the four images of each given cell.

S1-F1-Q5 · question page 16

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A figure looks the same after a reflection in a vertical line through its center and after a reflection in a horizontal line through its center. Then it also looks the same after a half-turn about its center.

Answer: True

Reasoning: Reflecting in the vertical line and then in the horizontal line sends (x, y) to (-x, y) and then to (-x, -y), which is a half-turn. Each reflection leaves the figure unchanged, so the half-turn does too.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S1-F2-Q1 · question page 17

LOGIC.SPATIAL.01 · Construct and count symmetric figures

The figure is on a 4 by 4 grid with a dashed vertical line between columns 2 and 3. Shaded cells can be added but not removed. What is the least number of cells you must add so that the figure looks the same in a mirror along the dashed line?

Answer: 5

Reasoning: For symmetry every shaded cell needs its image across the line (column c pairs with column 5 - c). Row 1: columns 1 and 2 need columns 4 and 3. Row 2: column 1 needs column 4; columns 2 and 3 are already each other's images. Row 3: column 1 needs column 4. Row 4: column 2 needs column 3. New cells: (1,3), (1,4), (2,4), (3,4), (4,3), so 5. Adding fewer leaves some shaded cell without its image.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-F2-Q2 · question page 18

LOGIC.SPATIAL.01 · Construct and count symmetric figures

Strings of 3 letters are made from the letters A, H, M, O and T (letters may repeat). Each is written left to right in block capitals. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. How many of these strings look exactly the same in a left-right mirror?

Answer: 25

Reasoning: All five letters are mirror-symmetric, so a string looks the same exactly when it reads the same backward: the first and last letters match. Choose the first letter (5 ways) and the middle letter (5 ways): 25.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-F2-Q3 · question page 19

LOGIC.SPATIAL.01 · Construct and count symmetric figures

Two vertical mirror lines are 3 cells apart: line 1 on the left, line 2 on the right. A shape is reflected in line 1 and the image is then reflected in line 2. Where does the final image end up?

Answer: D. the original shape moved 6 cells to the right, not flipped

Reasoning: Put line 1 at x = a and line 2 at x = b = a + 3. The first reflection sends x to 2a - x; the second sends that to 2b - (2a - x) = x + 2(b - a) = x + 6. Every point moves 6 to the right and nothing is flipped.

✗ the original shape moved 3 cells to the right, not flipped: two reflections move the shape twice the distance between the lines

✗ a mirror image of the shape, 6 cells to the right: two reflections undo each other's flip

✗ back exactly where the shape started: that only happens when both lines are the same line

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S1-F2-Q4 · question page 20

LOGIC.SPATIAL.01 · Construct and count symmetric figures

A row of 12 cells is numbered 1 to 12. A mirror line lies between cells 6 and 7. Give two cell numbers a and b, with a less than b, that are mirror images of each other.

Answer: any values that satisfy: a + b == 13 and a < b and a >= 1 and b <= 12 and isInteger(a) and isInteger(b) (example: a = 4, b = 9)

Reasoning: Cells that mirror each other are the same distance from the line, so their numbers add to 6 + 7 = 13: (1, 12), (2, 11), (3, 10), (4, 9), (5, 8) or (6, 7). Any one pair is correct.

Tempting answers the rule rejects: a = 4, b = 10; a = 6, b = 6; a = 9, b = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 6 valid answers.

S1-F2-Q5 · question page 21

LOGIC.SPATIAL.01 · Construct and count symmetric figures

Count only mirror lines that run vertically, horizontally or diagonally through the center of the shape (the middle of the smallest rectangle around it). Which figure has exactly ONE mirror line?

Answer: A. Figure A

Reasoning: Testing each kind of line: A (a T shape) looks the same only in the vertical line. B looks the same after a half-turn but in no mirror; C and D each have four mirror lines.

- ✗ Figure B: no mirror line: it only looks the same after a half-turn
- ✗ Figure C: four mirror lines (vertical, horizontal and both diagonals)
- ✗ Figure D: four mirror lines (vertical, horizontal and both diagonals)

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S2-M-Q1 · question page 23

LOGIC.READING.01 · Read negations, units and relationships exactly as stated

Three students each make one statement about themself, and every statement is true. Dee: "It is not true that I finished the race." Eli: "I did not fail to finish the race." Fay: "I never said that I finished the race." Who definitely finished the race?

Answer: B. Eli only

Reasoning: Eli: "did not fail to finish" is a double negative, so Eli finished. Dee denies finishing. Fay's statement is about what she said, not about the race, so she may or may not have finished. Only Eli definitely did.

- ✗ Dee only: "It is not true that I finished" means Dee did not finish
- ✗ Eli and Fay: Fay only says what she never SAID; that tells us nothing about whether she finished
- ✗ Dee and Fay: Dee did not finish, and nothing is known about Fay

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S2-M-Q2 · question page 24

LOGIC.READING.01 · Read negations, units and relationships exactly as stated

Shoelaces are sold only in packs of 2 laces for $3 a pack. Each sneaker needs exactly one

lace. A club needs laces for 12 sneakers and also buys 4 whistles at $5 each. What is the least the club can spend, in dollars?

Answer: 38

Reasoning: 12 sneakers need 12 laces = 6 packs, costing 6 x $3 = $18. Whistles cost 4 x $5 = $20. Total $38. A common slip is to pay $3 per lace (12 x 3 = $36, total $56).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-M-Q3 · question page 25

LOGIC.READING.01 · Read negations, units and relationships exactly as stated

In the Ortiz family, each son has exactly one brother, and each daughter has exactly two brothers and three sisters. How many children are in the family?

Answer: 6

Reasoning: A son's brothers are the other sons, so one brother means 2 sons. A daughter's sisters are the other daughters, so three sisters means 4 daughters, and each daughter's two brothers are the 2 sons. 2 + 4 = 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-M-Q4 · question page 26

LOGIC.READING.01 · Read negations, units and relationships exactly as stated

A box holds only red pens and blue pens: at least 3 red pens, at most 2 blue pens, and exactly 6 pens in all. Which statement MUST be true?

Answer: C. There are at least 4 red pens.

Reasoning: Blue is 0, 1 or 2, so red is 6, 5 or 4: always at least 4. Each

other statement fails in at least one of these three possible boxes.

- ✗ There are exactly 3 red pens.: then there would be 3 blue pens, but there are at most 2
- ✗ There are exactly 2 blue pens.: 1 or 0 blue pens also fit every condition
- ✗ There are no blue pens.: 1 or 2 blue pens also fit

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all (red, blue) pairs enumerated: (4, 2), (5, 1), (6, 0). Automatic: the check function accepts only the keyed choice.

S2-M-Q5 · question page 27

LOGIC.READING.01 · Read negations, units and relationships exactly as stated

On this team, April, June and May are the names of three players (not months). A coach writes: "June scored after April and before May." Put the players in the order in which they scored, first to last.

Answer: April → June → May

Reasoning: The words are names here, so read the sentence literally: April, then June, then May.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S2-B1-Q1 · question page 28

LOGIC.READING.01 · Strategy: rewrite each statement in plain form

Treat "unhappy" as meaning exactly "not happy". What does "Rui is not unhappy" say?

Answer: D. Rui is happy.

Reasoning: "Not unhappy" = not (not happy) = happy, under the stated meaning.

✗ Rui is unhappy.: the "not" cancels the "un-"

✗ Rui is sad and tired.: nothing is said about being tired

✗ Nothing at all about Rui.: it is a definite statement about Rui

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S2-B1-Q2 · question page 29

LOGIC.READING.01 · Strategy: rewrite each statement in plain form

A drawer holds 18 single socks that make complete matching pairs. How many pairs is that?

Answer: 9

Reasoning: A pair is 2 socks: 18 / 2 = 9.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-B1-Q3 · question page 30

LOGIC.READING.01 · Strategy: rewrite each statement in plain form

"At most 3 people may enter" allows exactly 3 people to enter.

Answer: True

Reasoning: "At most 3" means 3 or fewer, so 3 is allowed.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S2-B1-Q4 · question page 31

LOGIC.READING.01 · Strategy: rewrite each statement in plain form

"Not all of the doors are locked." Which statement means the same?

Answer: A. At least one door is unlocked.

Reasoning: "Not all are locked" is true exactly when at least one door is not locked.

- ✗ No door is locked.: that says more: some doors may still be locked
- ✗ Every door is unlocked.: that says more than the sentence
- ✗ Exactly one door is unlocked.: the sentence does not say how many

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S2-B1-Q5 · question page 32

LOGIC.READING.01 · Strategy: rewrite each statement in plain form

Lin has 3 brothers. Each of her brothers has exactly one sister. How many children are in the family?

Answer: 4

Reasoning: The brothers' one sister is Lin herself: 3 brothers + Lin = 4.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-F1-Q1 · question page 33

LOGIC.READING.01 · Read layered negations, units and family clues

Every statement below is true. Which person is going to the game? Gus: "It is not true that I am not going." Hal: "I am not not staying home." Ivy: "It is false that I won't go."

Answer: B. Gus and Ivy

Reasoning: Gus: not (not going) = going. Hal: not (not staying) = staying home. Ivy: false that (won't go) = will go.

✗ Gus only: Ivy's statement also cancels to "I will go"

✗ Hal and Ivy: Hal's "not not staying" means Hal is staying home

✗ all three: Hal is staying home

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S2-F1-Q2 · question page 34

LOGIC.READING.01 · Read layered negations, units and family clues

Eggs are sold in half-dozen boxes (6 eggs) for $2 a box. A recipe uses 4 eggs per cake, and a baker makes 9 cakes. What is the least she can spend on eggs, in dollars?

Answer: 12

Reasoning: 9 cakes x 4 eggs = 36 eggs = 6 boxes. 6 x $2 = $12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-F1-Q3 · question page 35

LOGIC.READING.01 · Read layered negations, units and family clues

In the Novak family, each son has exactly 3 sisters and 2 brothers. Each daughter has exactly 2 sisters. How many children are in the family?

Answer: 6

Reasoning: A son has 2 brothers, so there are 3 sons. A son has 3 sisters, so there are 3 daughters; then each daughter has 2 sisters, as stated. 3 + 3 = 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-F1-Q4 · question page 36

LOGIC.READING.01 · Read layered negations, units and family clues

A bag holds only red and blue marbles: at least twice as many red as blue, at least 3 blue, and at most 10 marbles in all. Which statement MUST be true?

Answer: C. There are exactly 3 blue marbles.

Reasoning: If blue were 4, red would be at least 8, giving at least 12 marbles: too many. So blue is exactly 3, and red is 6 or 7.

- ✗ There are exactly 6 red marbles.: 7 red and 3 blue also fit
- ✗ There are exactly 10 marbles.: 6 red and 3 blue (9 marbles) also fit
- ✗ There are more blue marbles than red ones.: red is at least twice blue

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: every (red, blue) pair up to 10 enumerated: (6, 3), (7, 3). Automatic: the check function accepts only the keyed choice.

S2-F1-Q5 · question page 37

LOGIC.READING.01 · Read layered negations, units and family clues

"Only members may vote" means that everyone who votes is a member.

Answer: True

Reasoning: "Only members may vote": if someone votes, that person is a member. It does not say every member votes.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S2-F2-Q1 · question page 38

LOGIC.READING.01 · Precise reading in multi-step puzzles

Four statements: (1) "It is not the case that 7 is not odd." (2) "It is false that 10 is not even."

(3) "It is not true that 9 is not a multiple of 4." (4) "5 is not unequal to 5." How many of the four statements are true?

Answer: 3

Reasoning: (1) 7 is odd: true. (2) 10 is even: true. (3) says 9 IS a multiple of 4: false. (4) "not unequal" means equal: 5 = 5, true. Three are true.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-F2-Q2 · question page 39

LOGIC.READING.01 · Precise reading in multi-step puzzles

On a quiz show, Monday, Tuesday, Wednesday and Friday are the first names of four contestants. The host says: "Wednesday answered before Monday. Friday answered right after Tuesday. Monday did not answer last." Put the contestants in answering order, first to last.

Answer: Wednesday → Monday → Tuesday → Friday

Reasoning: The words are names. Tuesday and Friday form a block (Tuesday, then Friday). With Wednesday before Monday, the possible orders are: block, Wednesday, Monday; Wednesday, block, Monday; Wednesday, Monday, block. Monday is not last, so only Wednesday, Monday, Tuesday, Friday remains.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S2-F2-Q3 · question page 40

LOGIC.READING.01 · Precise reading in multi-step puzzles

A bag holds only red and blue marbles: the number of red marbles is at least twice the number of blue, there are at least 3 blue marbles, and there are at most 10 marbles in all. Give a possible number of red marbles r and blue marbles b.

Answer: any values that satisfy: r >= 2*b and b >= 3 and r + b <= 10 and isInteger(r) and isInteger(b) (example: r = 6, b = 3)

Reasoning: Blue must be 3 (with 4 blue, red is at least 8, too many in all). Red can be 6 or 7. Either (6, 3) or (7, 3) is correct.

Tempting answers the rule rejects: r = 5, b = 3; r = 8, b = 4; r = 8, b = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 2 valid answers.

S2-F2-Q4 · question page 41

LOGIC.READING.01 · Precise reading in multi-step puzzles

In a group of brothers and sisters, every girl has exactly as many brothers as sisters, and there are 3 boys. How many girls are there?

Answer: 4

Reasoning: A girl's brothers are the 3 boys; her sisters are the other girls. So girls - 1 = 3, and there are 4 girls.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-F2-Q5 · question page 42

LOGIC.READING.01 · Precise reading in multi-step puzzles

"No student who did not study passed the test." Which statement follows?

Answer: A. Every student who passed studied.

Reasoning: The sentence says: if a student did not study, the student did not pass. Equivalently, if a student passed, the student studied (the

contrapositive).

- ✗ Every student who studied passed.: that reverses the condition
- ✗ Some students who studied failed.: nothing is said about students who studied
- ✗ No student passed.: students who studied may have passed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-M-Q1 · question page 44

LOGIC.ARGUMENT.01 · Identify statements, premises, conclusions and assumptions

"The trail is closed whenever it rains. It is raining now. So the trail is closed." Which sentence is the conclusion?

Answer: D. The trail is closed.

Reasoning: "So" introduces the conclusion: the trail is closed. The other two sentences are the reasons (premises).

- ✗ It is raining now.: this is given as a reason (a premise)
- ✗ The trail is closed whenever it rains.: this is a general premise
- ✗ Rain is likely tomorrow.: this sentence is not in the argument

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-M-Q2 · question page 45

LOGIC.ARGUMENT.01 · Identify statements, premises, conclusions and assumptions

Which of these is a statement (a sentence that is either true or false)?

Answer: B. Every square has four sides.

Reasoning: A statement makes a claim that is true or false. "Every

square has four sides" is a (true) statement.

- ✗ Please close the window.: a request is neither true nor false
- ✗ Is 15 a prime number?: a question is neither true nor false
- ✗ What a great game!: an exclamation of feeling is not a claim that is true or false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-M-Q3 · question page 46

LOGIC.ARGUMENT.01 · Identify statements, premises, conclusions and assumptions

"Since every ticket holder may enter, and Zoe holds a ticket, Zoe may enter." Which sentence is a premise?

Answer: C. Zoe holds a ticket.

Reasoning: "Since" introduces the reasons: every ticket holder may enter, and Zoe holds a ticket. The conclusion is that Zoe may enter.

- ✗ Zoe may enter.: this is the conclusion the argument reaches
- ✗ Zoe bought the ticket online.: the argument does not say this
- ✗ Only Zoe may enter.: the argument does not say this

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-M-Q4 · question page 47

LOGIC.ARGUMENT.01 · Identify statements, premises, conclusions and assumptions

"Ravi will be on time, because the bus left on schedule." Which unstated assumption does this argument need?

Answer: D. Ravi is on that bus.

Reasoning: The reason (the bus left on schedule) supports the

conclusion only if Ravi is riding it. That missing link is the assumption.

- ✗ The bus is red.: the color is irrelevant to the conclusion
- ✗ Ravi was late yesterday.: that would not help the conclusion
- ✗ Some buses are late.: that does not connect the bus to Ravi

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-M-Q5 · question page 48

LOGIC.ARGUMENT.01 · Identify statements, premises, conclusions and assumptions

"All of our club's robots passed inspection. Unit 7 is one of our club's robots. The room is cold. Therefore Unit 7 passed inspection." Which sentence plays no part in the reasoning?

Answer: A. The room is cold.

Reasoning: The conclusion follows from the first two sentences. The temperature of the room is irrelevant.

- ✗ Unit 7 is one of our club's robots.: this premise connects Unit 7 to the group
- ✗ All of our club's robots passed inspection.: this premise gives the general rule
- ✗ Therefore Unit 7 passed inspection.: this is the conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-B1-Q1 · question page 49

LOGIC.ARGUMENT.01 · Which sentences are statements?

"Close the door." is a statement.

Answer: False

Reasoning: It is a command; it is neither true nor false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-B1-Q2 · question page 50

LOGIC.ARGUMENT.01 · Which sentences are statements?

"7 is greater than 9." is a statement.

Answer: True

Reasoning: It is false, but it is still a statement: it makes a claim that is true or false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-B1-Q3 · question page 51

LOGIC.ARGUMENT.01 · Which sentences are statements?

Which is NOT a statement?

Answer: B. Who won the game?

Reasoning: A question asks; it does not claim.

- ✗ The game ended in a tie.: a claim that is true or false
- ✗ Nobody scored.: a claim that is true or false
- ✗ Our team won.: a claim that is true or false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-B1-Q4 · question page 52

LOGIC.ARGUMENT.01 · Which sentences are statements?

"x + 2 = 5" with no value of x given is a statement.

Answer: False

Reasoning: Its truth depends on x, which is not given; it becomes a statement once x is fixed (an open sentence).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-B1-Q5 · question page 53

LOGIC.ARGUMENT.01 · Which sentences are statements?

Which sentence is a statement?

Answer: C. Paris is in Spain.

Reasoning: It is a false claim, but a claim: a statement.

- ✗ Let's go to Paris!: a suggestion
- ✗ Is Paris far?: a question
- ✗ Visit Paris.: a command

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-B2-Q1 · question page 54

LOGIC.ARGUMENT.01 · Find the conclusion by its signal words

Which word usually introduces a conclusion?

Answer: B. therefore

Reasoning: "Therefore" signals the conclusion.

- ✗ because: introduces a reason
- ✗ since: introduces a reason

✗ given that: introduces a reason

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-B2-Q2 · question page 55

LOGIC.ARGUMENT.01 · Find the conclusion by its signal words

Which word usually introduces a reason (a premise)?

Answer: C. because

Reasoning: "Because" signals a reason.

✗ so: introduces a conclusion
✗ thus: introduces a conclusion
✗ consequently: introduces a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-B2-Q3 · question page 56

LOGIC.ARGUMENT.01 · Find the conclusion by its signal words

"The lake is frozen, so we can skate." What is the conclusion?

Answer: D. We can skate.

Reasoning: "So" introduces the conclusion.

✗ The lake is frozen.: the reason
✗ It is winter.: not stated
✗ Skating is fun.: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S3-B2-Q4 · question page 57

LOGIC.ARGUMENT.01 · Find the conclusion by its signal words

"We should leave now, since the show starts at 7." What is the conclusion?

Answer: A. We should leave now.

Reasoning: The claim supported by the "since" reason is the conclusion.

- ✗ The show starts at 7.: the reason, after "since"
- ✗ The show is long.: not stated
- ✗ We are late.: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S3-B2-Q5 · question page 58

LOGIC.ARGUMENT.01 · Find the conclusion by its signal words

In "Because the battery is dead, the car will not start," the conclusion is "the battery is dead."

Answer: False

Reasoning: "Because" marks the reason; the conclusion is that the car will not start.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-F1-Q1 · question page 59

LOGIC.ARGUMENT.01 · Arguments with several premises and hidden structure

"The museum must be open today. After all, it is open every weekday except Monday, and today is Thursday." Which sentence is the conclusion?

Answer: A. The museum must be open today.

Reasoning: The first sentence is supported by the other two ("after all" introduces reasons).

- ✗ Today is Thursday.: a premise
- ✗ It is open every weekday except Monday.: a premise
- ✗ The museum is closed on Monday.: part of a premise, not the claim being supported

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-F1-Q2 · question page 60

LOGIC.ARGUMENT.01 · Arguments with several premises and hidden structure

Put these sentences in the order of a clear argument: the general rule first, then the fact, then the conclusion.

Answer: Any whole number greater than 1 with a factor other than 1 and itself is not prime. → 91 = 7 x 13, so 91 has the factor 7. → Therefore, 91 is not prime.

Reasoning: Rule (B), fact about 91 (C), then the conclusion (A).

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S3-F1-Q3 · question page 61

LOGIC.ARGUMENT.01 · Arguments with several premises and hidden structure

"Coach Lee will pick Maya, because Coach Lee always picks the fastest runner." Which assumption is needed?

Answer: D. Maya is the fastest runner.

Reasoning: The rule talks about the fastest runner; the conclusion needs Maya to be that runner.

✗ Maya wants to be picked.: not needed for the conclusion
✗ Coach Lee likes Maya.: not what the argument relies on
✗ Maya ran yesterday.: irrelevant to being the fastest

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-F1-Q4 · question page 62

LOGIC.ARGUMENT.01 · Arguments with several premises and hidden structure

An argument can have a true conclusion even when one of its premises is false.

Answer: True

Reasoning: For example: "All fish can fly; a pigeon is a fish; so a pigeon can fly." The premises are false, the conclusion is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-F1-Q5 · question page 63

LOGIC.ARGUMENT.01 · Arguments with several premises and hidden structure

"If the alarm is set, it rings at 6. It did not ring at 6. Hence it was not set." How many premises does this argument have?

Answer: B. 2

Reasoning: Two premises (the conditional and the fact that it did not ring), and one conclusion after "hence".

✗ 1: there are two reasons before "hence"
✗ 3: the third sentence is the conclusion
✗ 0: the first two sentences are reasons

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-F2-Q1 · question page 64

LOGIC.ARGUMENT.01 · Evaluate structure and missing links

"All sprinters on the team train daily. So Kim trains daily." Which added premise makes the conclusion follow for certain?

Answer: C. Kim is a sprinter on the team.

Reasoning: The general premise is about sprinters on the team; Kim must be one of them.

- ✗ Kim trains often.: does not make Kim a sprinter on the team
- ✗ Some sprinters train daily.: the conclusion is about Kim, who is still not linked to the group
- ✗ Kim is on the team.: Kim might be on the team but not a sprinter

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-F2-Q2 · question page 65

LOGIC.ARGUMENT.01 · Evaluate structure and missing links

"Every number divisible by 6 is even. So 84 is even." Which added premise makes this valid?

Answer: D. 84 is divisible by 6.

Reasoning: The rule applies to numbers divisible by 6, so we need 84 to be one: 84 = 6 x 14.

- ✗ 84 is divisible by 4.: true, but the premise is about divisibility by 6
- ✗ 84 is greater than 6.: size does not matter
- ✗ Some even numbers are divisible by 6.: still says nothing about 84

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-F2-Q3 · question page 66

LOGIC.ARGUMENT.01 · Evaluate structure and missing links

In a valid argument, adding an extra true premise can make the conclusion no longer follow.

Answer: False

Reasoning: If the conclusion follows from some premises, it still follows when more premises are added.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-F2-Q4 · question page 67

LOGIC.ARGUMENT.01 · Evaluate structure and missing links

Arrange these sentences into an argument that ends with its conclusion: chain the two rules.

Answer: The dam opens. → If the dam opens, the river rises. → If the river rises, the ferry stops. → So the ferry stops.

Also accepted: If the dam opens, the river rises. → If the river rises, the ferry stops. → The dam opens. → So the ferry stops. | If the dam opens, the river rises. → The dam opens. → If the river rises, the ferry stops. → So the ferry stops.

Reasoning: State the first rule and the fact, then the second rule, then the conclusion. Orders that state both rules first are also accepted.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 3 valid.

S3-F2-Q5 · question page 68

LOGIC.ARGUMENT.01 · Evaluate structure and missing links

"Our new ads worked: sales rose this month." Which assumption is the argument making?

Answer: B. Nothing other than the ads caused most of the rise in sales.

Reasoning: Claiming the ads "worked" assumes the rise was due to the ads and not to something else.

- ✗ Sales rose this month.: that is the stated premise, not an assumption
- ✗ The ads were expensive.: cost is irrelevant to whether they worked
- ✗ Sales will rise next month.: the argument is about this month

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S4-M-Q1 · question page 70

LOGIC.CONSTRAINT.01 · Find the only candidate left after every clue

Which month is it? Its name has fewer than 8 letters. Its name contains the letter R. It is the 1st, 3rd, 5th, 7th, 9th or 11th month. Its name does not start with J.

Answer: A. March

Reasoning: Fewer than 8 letters and an R: January, March, April, October. Odd-numbered month (1st, 3rd, ...): January, March. Not starting with J: March.

- ✗ January: starts with J
- ✗ May: has no R
- ✗ September: has 9 letters

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the four clues applied to all 12 months leave only March. Automatic: the check function accepts only the keyed choice.

S4-M-Q2 · question page 71

LOGIC.CONSTRAINT.01 · Find the only candidate left after every clue

Which day of the week is it? Its name has no letter E. It comes after Wednesday in the week that starts on Monday. Its name has exactly 6 letters. Its name contains the letter U.

Answer: C. Sunday

Reasoning: After Wednesday: Thursday, Friday, Saturday, Sunday (none has an E). Exactly 6 letters: Friday, Sunday. Contains a U: Sunday.

- ✗ Friday: fits the first three clues but has no letter U
- ✗ Saturday: has 8 letters
- ✗ Thursday: has 8 letters

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the four clues applied to all 7 days leave only Sunday. Automatic: the check function accepts only the keyed choice.

S4-M-Q3 · question page 72

LOGIC.CONSTRAINT.01 · Find the only candidate left after every clue

A whole number from 1 to 50 leaves remainder 3 when divided by 7. It is greater than 20. It is even. It does not leave remainder 4 when divided by 5. What is the number?

Answer: 38

Reasoning: Remainder 3 mod 7: 3, 10, 17, 24, 31, 38, 45. Greater than 20: 24, 31, 38, 45. Even: 24, 38. 24 leaves remainder 4 when divided by 5, so the number is 38.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S4-M-Q4 · question page 73

LOGIC.CONSTRAINT.01 · Find the only candidate left after every clue

Three friends, Ana, Ben and Cai, each have exactly one pet: a cat, a dog or a fish, all different. Ana does not have the dog. Ben has neither the cat nor the dog. Which pet does

each friend have?

Answer: Ana: cat; Ben: fish; Cai: dog

Reasoning: Ben has neither cat nor dog, so Ben has the fish. Ana does not have the dog, so Ana has the cat, and Cai has the dog.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S4-M-Q5 · question page 74

LOGIC.CONSTRAINT.01 · Find the only candidate left after every clue

A locker number is one of 12, 15, 18, 21, 24, 27. It is even. It is not a multiple of 4. Which locker is it?

Answer: D. 18

Reasoning: Even: 12, 18, 24. Not a multiple of 4: 18.

- ✗ 27: 27 is odd, so it is not even
- ✗ 12: a multiple of 4
- ✗ 24: a multiple of 4

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S4-B1-Q1 · question page 75

LOGIC.CONSTRAINT.01 · Strategy: cross out, one clue at a time

A number from 1 to 10 is even, greater than 7 and less than 10. What is it?

Answer: 8

Reasoning: Even: 2, 4, 6, 8, 10. Greater than 7: 8, 10. Less than 10: 8.

Grading: The game's answer checker compares the typed value with the key (equivalent

number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S4-B1-Q2 · question page 76

LOGIC.CONSTRAINT.01 · Strategy: cross out, one clue at a time

A day of the week (Monday to Sunday) starts with T and has more than 7 letters. Which day?

Answer: B. Thursday

Reasoning: T-days: Tuesday (7), Thursday (8).

- ✗ Tuesday: has exactly 7 letters
- ✗ Saturday: does not start with T
- ✗ Wednesday: does not start with T

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S4-B1-Q3 · question page 77

LOGIC.CONSTRAINT.01 · Strategy: cross out, one clue at a time

How many months have a name with exactly 4 letters?

Answer: 2

Reasoning: June and July.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S4-B1-Q4 · question page 78

LOGIC.CONSTRAINT.01 · Strategy: cross out, one clue at a time

If two candidates are left after every clue, the clues determine the answer.

Answer: False

Reasoning: Two candidates fit every clue, so the answer is not determined.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S4-B1-Q5 · question page 79

LOGIC.CONSTRAINT.01 · Strategy: cross out, one clue at a time

Candidates: 2, 3, 4, 5, 6. Clue 1: the number is prime. Clue 2: the number is odd. How many candidates are left?

Answer: C. 2

Reasoning: Primes: 2, 3, 5. Odd: 3, 5.

- ✗ 1: 3 and 5 are both odd primes
- ✗ 3: 2 is prime but even
- ✗ 0: 3 and 5 fit

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S4-F1-Q1 · question page 80

LOGIC.CONSTRAINT.01 · Eliminate with clues about neighbors and positions

Which month is it? The month before it has a name longer than 5 letters. The month after it does not end in "ber". The month after it contains the letter U. Its own name contains the letter A.

Answer: A. January

Reasoning: Months whose previous month has more than 5 letters:

January, February, March, September, October, November, December.
Next month not ending in "ber": January, February, March, December.
Next month containing U: January (February) and December (January).
Own name with an A: January.

- ✗ December: fits the three neighbor clues but has no letter A
- ✗ February: the month after it, March, has no letter U
- ✗ May: the month before it, April, has only 5 letters

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 12 months tested against the four neighbor clues. Automatic: the check function accepts only the keyed choice.

S4-F1-Q2 · question page 81

LOGIC.CONSTRAINT.01 · Eliminate with clues about neighbors and positions

In the week Monday to Sunday, exactly one day has a name whose day before and day after both contain the letter T.

Answer: False

Reasoning: Two days qualify: Wednesday (Tuesday and Thursday) and Friday (Thursday and Saturday).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S4-F1-Q3 · question page 82

LOGIC.CONSTRAINT.01 · Eliminate with clues about neighbors and positions

A whole number from 1 to 100 is a perfect square, is odd, has two digits, and its digits add to 13. What is it?

Answer: 49

Reasoning: Odd two-digit squares: 25, 49, 81. Digit sums 7, 13, 9. So 49.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S4-F1-Q4 · question page 83

LOGIC.CONSTRAINT.01 · Eliminate with clues about neighbors and positions

Four runners, Dev, Eva, Finn and Gia, finished 1st to 4th, no ties. Eva finished right after Dev. Gia finished before Finn. Finn finished before Dev. Assign each runner a place.

Answer: Dev: 3rd; Eva: 4th; Finn: 2nd; Gia: 1st

Reasoning: Gia is before Finn and Finn before Dev, so Dev is at least 3rd; Eva is right after Dev, so Dev is 3rd and Eva 4th. Then Gia is 1st and Finn 2nd.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S4-F1-Q5 · question page 84

LOGIC.CONSTRAINT.01 · Eliminate with clues about neighbors and positions

Candidates are the whole numbers 1 to 30. Clues: A: the number is odd. B: it is a multiple of 3. C: it is greater than 10. D: it is a multiple of 5. Which number fits all four clues?

Answer: B. 15

Reasoning: Odd multiples of 15 up to 30: 15 only.

- ✗ 30: even, so it fails clue A
- ✗ 5: not a multiple of 3 and not greater than 10
- ✗ 25: not a multiple of 3

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S4-F2-Q1 · question page 85

LOGIC.CONSTRAINT.01 · Which clues are really needed?

Candidates are the whole numbers 1 to 30, with four clues: A: odd; B: a multiple of 3; C: greater than 10; D: a multiple of 5. Only 15 fits all four. Which ONE clue can be removed so that exactly one number is still left?

Answer: A. C (greater than 10)

Reasoning: Remove each clue in turn and recount. Without C the clues say: odd, multiple of 3 and multiple of 5, that is an odd multiple of 15, and the only one up to 30 is 15. Removing any other clue leaves two or more numbers.

- ✗ B (multiple of 3): without B, 15 and 25 both fit
- ✗ A (odd): without A, 15 and 30 both fit
- ✗ D (multiple of 5): without D, 15, 21 and 27 fit

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: each clue removed in turn and the survivors recounted. Automatic: the check function accepts only the keyed choice.

S4-F2-Q2 · question page 86

LOGIC.CONSTRAINT.01 · Which clues are really needed?

A number from 1 to 30 is a multiple of 4. What is the least number of extra yes/no questions of the form "Is it greater than k?" that always identify it?

Answer: 3

Reasoning: Multiples of 4 up to 30: 4, 8, 12, 16, 20, 24, 28 (7 candidates). Each question at best halves the list, so 2 questions separate at most 4; 3 questions separate up to 8, and halving 7 -> 4 -> 2 -> 1 works.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S4-F2-Q3 · question page 87

LOGIC.CONSTRAINT.01 · Which clues are really needed?

Adding a new clue to a puzzle can never increase the number of candidates left.

Answer: True

Reasoning: Each clue only removes candidates; the survivors of more clues are a subset of the survivors of fewer.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S4-F2-Q4 · question page 88

LOGIC.CONSTRAINT.01 · Which clues are really needed?

Candidates are the whole numbers 1 to 20. Write a number n that survives BOTH clues: n is prime, and n leaves remainder 1 when divided by 4.

Answer: any values that satisfy: isPrime(n) and n % 4 == 1 and n >= 1 and n <= 20 (example: n = 5)

Reasoning: Primes up to 20 leaving remainder 1 mod 4: 5, 13, 17. Any one is correct.

Tempting answers the rule rejects: n = 9; n = 7; n = 21

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 3 valid answers.

S4-F2-Q5 · question page 89

LOGIC.CONSTRAINT.01 · Which clues are really needed?

Six boxes are numbered 1 to 6. One holds a prize. Which single clue, added alone, would leave exactly one box?

Answer: D. "The box number is prime and even."

Reasoning: The only even prime is 2.

- ✗ "The box number is odd.": leaves 1, 3, 5
- ✗ "The box number is less than 4.": leaves 1, 2, 3
- ✗ "The box number is a multiple of 3.": leaves 3 and 6

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S5-M-Q1 · question page 91

LOGIC.CONNECTIVE.01 · Form negations and find truth values

What is the negation of "Sam is taller than Ana"?

Answer: C. Sam is not taller than Ana.

Reasoning: A negation is true exactly when the original is false. "Sam is taller" is false when Sam is shorter OR the same height; only "Sam is not taller than Ana" covers both.

- ✗ Ana is taller than Sam.: this leaves out the case where they are the same height
- ✗ Sam is shorter than Ana.: this also leaves out equal heights
- ✗ Sam and Ana are the same height.: this leaves out the case where Ana is taller

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-M-Q2 · question page 92

LOGIC.CONNECTIVE.01 · Form negations and find truth values

x stands for a real number. What is the negation of "x > 5"?

Answer: A. x ≤ 5

Reasoning: "x > 5" is false exactly when x is 5 or less: x ≤ 5.

- ✗ x < 5: misses x = 5, where "x > 5" is false
- ✗ x ≥ 5: includes values like 6 that make "x > 5" true
- ✗ x < -5: misses every x from -5 to 5

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S5-M-Q3 · question page 93

LOGIC.CONNECTIVE.01 · Form negations and find truth values

P is the statement "48 is a multiple of 9." The negation of P is true.

Answer: True

Reasoning: 48 = 9 x 5 + 3, so P is false and its negation, "48 is not a multiple of 9", is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S5-M-Q4 · question page 94

LOGIC.CONNECTIVE.01 · Form negations and find truth values

Give a number x that makes the NEGATION of the statement "x ≤ 4" true.

Answer: any values that satisfy: x > 4 (example: x = 7)

Reasoning: The negation of "x ≤ 4" is "x > 4", so any number greater than 4 is correct.

Tempting answers the rule rejects: x = 4; x = 3; x = -5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S5-M-Q5 · question page 95

LOGIC.CONNECTIVE.01 · Form negations and find truth values

Which pair of statements are negations of each other (in every situation, exactly one of the two is true)?

Answer: C. "The coin landed heads up." / "The coin did not land heads up."

Reasoning: Negations can never be both false or both true. Each other pair has a situation where both are false: they are opposites (contraries), not negations.

- ✗ "The cup is full." / "The cup is empty.": a half-full cup makes both false
- ✗ "Every student passed." / "Every student failed.": if some passed and some failed, both are false
- ✗ "x > 3" / "x < 3": x = 3 makes both false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-B1-Q1 · question page 96

LOGIC.ARGUMENT.01 · Truth values of simple statements

"17 is a prime number" is true.

Answer: True

Reasoning: 17 has no factors other than 1 and itself.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S5-B1-Q2 · question page 97

LOGIC.ARGUMENT.01 · Truth values of simple statements

"3 x 7 = 24" is true.

Answer: False

Reasoning: 3 x 7 = 21.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S5-B1-Q3 · question page 98

LOGIC.ARGUMENT.01 · Truth values of simple statements

"Every even number is a multiple of 4" is true.

Answer: False

Reasoning: 6 is even but not a multiple of 4, so the statement is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-B1-Q4 · question page 99

LOGIC.ARGUMENT.01 · Truth values of simple statements

Which statement is false?

Answer: D. 0.5 is greater than 0.75.

Reasoning: 0.5 < 0.75.

✗ 0.5 equals one half.: this is true: 0.5 = 1/2

✗ 0.75 is greater than 0.5.: this is true: 75 hundredths > 50 hundredths
✗ 0.5 + 0.25 = 0.75.: this is true: 50 + 25 = 75 hundredths

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-B1-Q5 · question page 100

LOGIC.ARGUMENT.01 · Truth values of simple statements

"10 is not an even number" is true.

Answer: False

Reasoning: 10 = 2 x 5 is even, so "10 is not even" is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-B2-Q1 · question page 101

LOGIC.CONNECTIVE.01 · Negation is not the same as the opposite

What is the negation of "The door is open"?

Answer: B. The door is not open.

Reasoning: Put "not" on the whole claim.

✗ The door is locked.: a door can be closed without being locked
✗ The window is open.: that is about the window
✗ The door is wide open.: that is not a negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-B2-Q2 · question page 102

LOGIC.CONNECTIVE.01 · Negation is not the same as the opposite

"The water is cold" and "The water is hot" are negations of each other.

Answer: False

Reasoning: Warm water makes both false; they are opposites, not negations.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-B2-Q3 · question page 103

LOGIC.CONNECTIVE.01 · Negation is not the same as the opposite

What is the negation of "Tom won the race"?

Answer: A. Tom did not win the race.

Reasoning: Negate the whole statement.

- ✗ Tom lost the race.: he may not have raced at all
- ✗ Tom came last.: too specific
- ✗ Someone else won.: there may have been no winner

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-B2-Q4 · question page 104

LOGIC.CONNECTIVE.01 · Negation is not the same as the opposite

If a statement is true, its negation is also true.

Answer: False

Reasoning: A statement and its negation always have opposite truth values.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-B2-Q5 · question page 105

LOGIC.CONNECTIVE.01 · Negation is not the same as the opposite

n is a whole number. What is the negation of "n is even"?

Answer: C. n is odd.

Reasoning: For whole numbers, "not even" means exactly "odd".

- ✗ n is 0.: too specific
- ✗ n is negative.: unrelated
- ✗ n is prime.: unrelated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-F1-Q1 · question page 106

LOGIC.CONNECTIVE.01 · Negate ranges, counts and double negatives

x stands for a real number. What is the negation of "$3 < x < 7$" (x is between 3 and 7)?

Answer: D. $x \leq 3$ or $x \geq 7$

Reasoning: x is not strictly between 3 and 7 exactly when it is at most 3 or at least 7.

- ✗ $x < 3$ or $x > 7$: misses $x = 3$ and $x = 7$
- ✗ $3 \geq x \geq 7$: no number satisfies this
- ✗ $x \leq 3$ and $x \geq 7$: no number satisfies this

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S5-F1-Q2 · question page 107

LOGIC.CONNECTIVE.01 · Negate ranges, counts and double negatives

Exactly 9 lamps are in a room. What is the negation of "At least 5 lamps are on"?

Answer: B. At most 4 lamps are on.

Reasoning: With whole numbers 0 to 9, "at least 5 on" is false exactly when 0 to 4 are on: at most 4.

- ✗ At most 5 lamps are on.: exactly 5 lamps on makes both statements true
- ✗ No lamps are on.: too strong: 1 to 4 lamps on also makes the original false
- ✗ At least 4 lamps are off.: with 5 on and 4 off, the original and this statement are both true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S5-F1-Q3 · question page 108

LOGIC.CONNECTIVE.01 · Negate ranges, counts and double negatives

Complete the truth table: the columns are "not P" and "not (not P)".

Answer: F T / T F

Reasoning: "not P" flips each value; flipping again gives back P.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (2 rows).

S5-F1-Q4 · question page 109

LOGIC.CONNECTIVE.01 · Negate ranges, counts and double negatives

How many of these are true? (1) "It is not true that 8 is odd." (2) "It is false that 0 is not a whole number." (3) "It is not the case that 2 + 2 is not 4." (4) "It is not true that 5 is not less

than 3."

Answer: 3

Reasoning: (1) 8 is not odd: true. (2) says 0 IS a whole number: true. (3) says 2 + 2 is 4: true. (4) says 5 IS less than 3: false. Three are true.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S5-F1-Q5 · question page 110

LOGIC.CONNECTIVE.01 · Negate ranges, counts and double negatives

The negation of "Mia always walks to school" is "Mia never walks to school".

Answer: False

Reasoning: If Mia walks on some days and not others, both are false. The negation is "Mia does not always walk to school": on at least one day she does not walk.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-F2-Q1 · question page 111

LOGIC.CONNECTIVE.01 · Negation inside conditions and counts

Give a number x that makes the negation of "x < 2 or x > 5" true.

Answer: any values that satisfy: x >= 2 and x <= 5 (example: x = 3)

Reasoning: "x < 2 or x > 5" is false exactly when x is not less than 2 AND not greater than 5: 2 ≤ x ≤ 5.

Tempting answers the rule rejects: x = 1; x = 6; x = 5.5

Grading: The learner's values are put into the rule; ANY values that make the rule true are

correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S5-F2-Q2 · question page 112

LOGIC.CONNECTIVE.01 · Negation inside conditions and counts

There are 6 lights. What is the negation of "At most 2 lights are on"?

Answer: C. At least 3 lights are on.

Reasoning: "At most 2" is false exactly when 3 or more are on.

- ✗ At least 2 lights are on.: exactly 2 on makes both true
- ✗ Exactly 3 lights are on.: too narrow: 4, 5 or 6 on also make the original false
- ✗ At most 2 lights are off.: 3 lights on and 3 off makes both false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S5-F2-Q3 · question page 113

LOGIC.CONNECTIVE.01 · Negation inside conditions and counts

Ana says: "It is false that it is not true that the test is not on Friday." What is Ana saying?

Answer: D. The test is not on Friday.

Reasoning: Count: "false that" (1), "not true that" (2), "not on Friday" (3). Two negations cancel, leaving one: the test is not on Friday.

- ✗ The test is on Friday.: count the negations: there are three
- ✗ Nothing about the test.: it is a definite claim
- ✗ The test might be on Friday.: the statement is definite

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-F2-Q4 · question page 114

LOGIC.CONNECTIVE.01 · Negation inside conditions and counts

For every statement P, exactly one of P and "not P" is true.

Answer: True

Reasoning: That is what a negation means: it is true exactly when P is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-F2-Q5 · question page 115

LOGIC.CONNECTIVE.01 · Negation inside conditions and counts

Statements about a whole number n from 1 to 10: A: "n is even." B: "n is not greater than 6." C: "It is not true that n is a multiple of 3." For how many values of n are all three statements true?

Answer: 2

Reasoning: Even and at most 6: 2, 4, 6. Not a multiple of 3: 2, 4. Two values.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-M-Q1 · question page 117

LOGIC.CODE.01 · Decode and encode with a stated key, and find a key within a stated family

In a code, each letter has the value of its position in the alphabet (A = 1, B = 2, ..., Z = 26), and a word's value is the sum of its letters' values. For example, CAB = 3 + 1 + 2 = 6. What

is the value of BEAD?

Answer: 12

Reasoning: B = 2, E = 5, A = 1, D = 4: 2 + 5 + 1 + 4 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-M-Q2 · question page 118

LOGIC.CODE.01 · Decode and encode with a stated key, and find a key within a stated family

In a code, every letter is replaced by the letter 3 places later in the alphabet (A becomes D, B becomes E, and X, Y, Z wrap around to A, B, C). A coded word reads FDW. What was the original word?

Answer: B. CAT

Reasoning: Decoding moves each letter 3 places back: F -> C, D -> A, W -> T.

- ✗ IGZ: that shifts 3 more places forward instead of back
- ✗ CAW: only the first two letters were decoded
- ✗ DBX: each letter was moved back only 2 places

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S6-M-Q3 · question page 119

LOGIC.CODE.01 · Decode and encode with a stated key, and find a key within a stated family

A code adds the same whole number k to every digit and keeps only the last digit of each sum (so 8 + 3 gives 1). The code turns 2025 into 4247. What does it turn 1990 into?

Answer: 3112

Reasoning: From 2 -> 4, k = 2 (every digit agrees: 0 -> 2, 2 -> 4, 5 -> 7). Then 1 -> 3, 9 -> 1, 9 -> 1, 0 -> 2: 3112.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-M-Q4 · question page 120

LOGIC.CODE.01 · Decode and encode with a stated key, and find a key within a stated family

Each symbol stands for a whole number. Star + Star = 10. Star + Circle = 8. Circle + Square = 9. What number is Square?

Answer: 6

Reasoning: Star = 5, so Circle = 3, so Square = 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-M-Q5 · question page 121

LOGIC.CODE.01 · Decode and encode with a stated key, and find a key within a stated family

Two 3-letter words were interleaved: first letter of word 1, first letter of word 2, second letter of word 1, and so on. The result is CDAORT. Which two words were interleaved?

Answer: A. CAR and DOT

Reasoning: Letters 1, 3, 5 (C, A, R) form word 1 and letters 2, 4, 6 (D, O, T) form word 2.

✗ COD and ART: interleaving COD and ART gives CAORDT

✗ CAT and DOR: interleaving gives CDAOTR
✗ CAD and ORT: interleaving gives COARDT

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S6-B1-Q1 · question page 122

LOGIC.CODE.01 · Strategy: write the key as a table

A code replaces each letter by its position in the alphabet (A = 1, ..., Z = 26) and adds them. What is the value of DOG?

Answer: 26

Reasoning: D = 4, O = 15, G = 7: 26.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-B1-Q2 · question page 123

LOGIC.CODE.01 · Strategy: write the key as a table

A code moves each letter 1 place later (A -> B, ..., Z -> A). What is BOX in code?

Answer: A. CPY

Reasoning: B -> C, O -> P, X -> Y.

✗ ANW: moved back instead of forward
✗ CPX: the last letter was not coded
✗ DQZ: moved 2 places

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S6-B1-Q3 · question page 124

LOGIC.CODE.01 · Strategy: write the key as a table

A code replaces each digit d by 9 - d. What is the code for 306?

Answer: 693

Reasoning: 3 -> 6, 0 -> 9, 6 -> 3.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-B1-Q4 · question page 125

LOGIC.CODE.01 · Strategy: write the key as a table

If a code moves every letter 2 places later, then decoding moves every letter 2 places earlier.

Answer: True

Reasoning: Decoding undoes encoding.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S6-B1-Q5 · question page 126

LOGIC.CODE.01 · Strategy: write the key as a table

Triangle + Triangle + Triangle = 12. What number is Triangle?

Answer: 4

Reasoning: 3 x 4 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F1-Q1 · question page 127

LOGIC.CODE.01 · Codes with structure: shapes of letters and combined rules

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. What is the value of BREAD?

Answer: 5

Reasoning: B 2 + R 1 + E 0 + A 1 + D 1 = 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F1-Q2 · question page 128

LOGIC.CODE.01 · Codes with structure: shapes of letters and combined rules

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. Which word has value 0?

Answer: B. FLIES

Reasoning: F, L, I, E and S all have 0.

- ✗ TOWEL: O has 1
- ✗ MAZE: A has 1
- ✗ SHIRT: R has 1

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S6-F1-Q3 · question page 129

LOGIC.CODE.01 · Codes with structure: shapes of letters and combined rules

A code first writes a word backward and then moves each letter 1 place later in the alphabet. What is the code for MAP?

Answer: QBN

Reasoning: Backward: PAM. Shift by 1: QBN.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F1-Q4 · question page 130

LOGIC.CODE.01 · Codes with structure: shapes of letters and combined rules

Each symbol stands for a whole number. Moon + Moon + Sun = 13. Sun + Sun = 6. Moon + Star = 9. What number is Star?

Answer: 4

Reasoning: Sun = 3, so 2 x Moon = 10 and Moon = 5, so Star = 4.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F1-Q5 · question page 131

LOGIC.CODE.01 · Codes with structure: shapes of letters and combined rules

A code moves every letter 13 places later in the alphabet (wrapping around). Applying this code twice gives back the original word.

Answer: True

Reasoning: 13 + 13 = 26 places, a full trip around the alphabet.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S6-F2-Q1 · question page 132

LOGIC.CODE.01 · Find a key and prove when none exists

A code adds 3 to each digit of a 4-digit number and keeps only the last digit of each sum (so 8 becomes 1). Is there a 4-digit number whose code is the same number written backward?

Answer: C. No: the middle digits would need b + 6 to end in the same digit as b, which never happens.

Reasoning: If abcd codes to dcba, then b + 3 ends in c and c + 3 ends in b, so b + 6 ends in b, which is impossible (6 is not a multiple of 10).

- ✗ Yes: 1234 works.: 1234 codes to 4567, not 4321
- ✗ Yes: any number whose digits are all equal works.: 5555 codes to 8888, not 5555
- ✗ No: but only because the first digit cannot be 0.: the reason is the middle digits, not a leading zero

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 9000 four-digit numbers tested; none works. Automatic: the check function accepts only the keyed choice.

S6-F2-Q2 · question page 133

LOGIC.CODE.01 · Find a key and prove when none exists

A code moves every letter the same number of places later in the alphabet (wrapping around). It turns GO into MU. What does it turn LOGIC into?

Answer: RUMOI

Reasoning: G -> M is 6 places, and O -> U is also 6. Shift LOGIC by 6: RUMOI.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F2-Q3 · question page 134

LOGIC.CODE.01 · Find a key and prove when none exists

Each symbol stands for a whole number from 1 to 9. Square + Square = Circle, and Circle + Square = 12. Find Square s and Circle c.

Answer: any values that satisfy: s + s == c and c + s == 12 and isInteger(s) and isInteger(c) (example: s = 4, c = 8)

Reasoning: 3 x Square = 12, so Square = 4 and Circle = 8. This is the only solution.

Tempting answers the rule rejects: s = 6, c = 6; s = 3, c = 9

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S6-F2-Q4 · question page 135

LOGIC.CODE.01 · Find a key and prove when none exists

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. Using only the letters B, O and T (each as often as you like), what is the greatest value of a 4-letter string?

Answer: 8

Reasoning: B has the most closed spaces (2), so BBBB = 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F2-Q5 · question page 136

LOGIC.CODE.01 · Find a key and prove when none exists

A code sends each letter to the letter at the same distance from the other end of the alphabet (A <-> Z, B <-> Y, C <-> X, ...). Which word is coded as itself?

Answer: D. No word: no letter is coded as itself.

Reasoning: A letter at position p codes to position 27 - p. A letter would code to itself only if p = 27 - p, which needs p = 13.5: impossible. So no word codes to itself.

- ✗ MOON: M codes to N and O to L
- ✗ WIZARD: W -> D, I -> R, ...: it codes to DRAZIW
- ✗ NOON: N codes to M

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S7-M-Q1 · question page 138

LOGIC.ORDER.01 · Build an ordering that satisfies relational clues

Four books stand on a shelf. The atlas is left of the cookbook. The diary is right of the cookbook. The biography is at the far left. Order the books from left to right.

Answer: biography → atlas → cookbook → diary

Reasoning: The biography is first. The atlas, cookbook and diary must come in that order after it.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S7-M-Q2 · question page 139

LOGIC.ORDER.01 · Build an ordering that satisfies relational clues

Five friends arrived at a party one at a time. Gil arrived before Hana but after Ivo. Jo arrived last. Kai arrived right after Ivo. Order the arrivals, first to last.

Answer: Ivo → Kai → Gil → Hana → Jo

Reasoning: Jo is last. Ivo comes before Gil, who comes before Hana, and Kai is right after Ivo: Ivo, Kai, Gil, Hana, Jo.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 120 orderings of 5 items enumerated; 1 valid.

S7-M-Q3 · question page 140

LOGIC.ORDER.01 · Build an ordering that satisfies relational clues

Five runners finished a race with no ties. Uma beat Vic. Wes finished right after Uma. Xia beat Uma. Yan finished last. Who finished in the middle (3rd)?

Answer: C. Wes

Reasoning: Yan is 5th. Xia beat Uma and Uma beat Vic, with Wes right after Uma: Xia, Uma, Wes, Vic, Yan.

- ✗ Uma: Uma is 2nd
- ✗ Vic: Vic is 4th
- ✗ Xia: Xia is 1st

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 120 orders checked; only Xia, Uma, Wes, Vic, Yan fits. Automatic: the check function accepts only the keyed choice.

S7-M-Q4 · question page 141

LOGIC.ORDER.01 · Build an ordering that satisfies relational clues

Four flags hang on a pole, from top to bottom. The green flag is somewhere between the red and the blue flags. The red flag is not at the top. The yellow flag is directly below the blue flag. Order the flags from top to bottom.

Answer: blue → yellow → green → red

Reasoning: Blue must be above yellow with nothing between them, and

green lies between red and blue. If blue were below green, yellow would follow blue at the bottom and red would have to be on top, which is not allowed. So blue is on top, then yellow, then green, then red.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S7-M-Q5 · question page 142

LOGIC.ORDER.01 · Build an ordering that satisfies relational clues

Seven students stand in a line. Lee is 2nd from the left. Max is 3rd from the right. How many students stand between Lee and Max?

Answer: 2

Reasoning: Max is 3rd from the right of 7, so 5th from the left. Between positions 2 and 5 are positions 3 and 4: 2 students.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S7-B1-Q1 · question page 143

LOGIC.ORDER.01 · Strategy: place the most constrained item first

Three cups stand in a row. The red cup is between the other two. The blue cup is on the right. Order them from left to right.

Answer: white → red → blue

Reasoning: Blue on the right, red in the middle, white on the left.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S7-B1-Q2 · question page 144

LOGIC.ORDER.01 · Strategy: place the most constrained item first

In a line of 9 people, Ola is 4th from the front. What is her position counted from the back?

Answer: 6

Reasoning: 9 - 4 + 1 = 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S7-B1-Q3 · question page 145

LOGIC.ORDER.01 · Strategy: place the most constrained item first

"Bea is right after Cy" means no one is between Cy and Bea.

Answer: True

Reasoning: "Right after" means directly next.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S7-B1-Q4 · question page 146

LOGIC.ORDER.01 · Strategy: place the most constrained item first

Ava is taller than Ben, and Ben is taller than Cal. Who is shortest?

Answer: B. Cal

Reasoning: Ava > Ben > Cal.

- ✗ Ava: Ava is tallest
- ✗ Ben: Ben is in the middle
- ✗ It cannot be known.: the clues give a full order

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S7-B1-Q5 · question page 147

LOGIC.ORDER.01 · Strategy: place the most constrained item first

Four racers: Dot finished first. Eli beat Fay. Gus finished last. Order them, first to last.

Answer: Dot → Eli → Fay → Gus

Reasoning: Dot first, Gus last, Eli before Fay.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S7-F1-Q1 · question page 148

LOGIC.ORDER.01 · Orderings with "next to" and "not next to"

Five houses stand in a row, numbered 1 to 5 from left to right. The bakery is next to the bank. The café is at one end. The dentist is directly right of the café. The florist is not next to the dentist. The bank is in house 4. Order the businesses from left to right.

Answer: café → dentist → bakery → bank → florist

Reasoning: The dentist is right of the café, so the café is house 1 and the dentist house 2. The bank is house 4. The florist is not next to the dentist, so not house 3: the florist is house 5 and the bakery house 3, next to the bank.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 120 orderings of 5 items enumerated; 1 valid.

S7-F1-Q2 · question page 149

LOGIC.ORDER.01 · Orderings with "next to" and "not next to"

Six people, A to F, sit in seats 1 to 6 in a row. C is in seat 1 and D is in seat 6. B sits right after A, and E sits right after B. F does not sit next to C. Which seat is F in?

Answer: 5

Reasoning: A, B, E form a block of three in seats 2-4 or 3-5. If the block were in 3-5, F would be in seat 2, next to C. So the block is in seats 2-4 and F is in seat 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S7-F1-Q3 · question page 150

LOGIC.ORDER.01 · Orderings with "next to" and "not next to"

Five students line up. Pat is in front of Quinn. Rae is right behind Sol. Tia is first. Quinn is last. Which statement must be true?

Answer: D. Tia is directly in front of Pat or of Sol.

Reasoning: Tia is 1st and Quinn 5th. The middle three are Pat and the block Sol-Rae in some order: Pat, Sol, Rae or Sol, Rae, Pat. In both, position 2 is Pat or Sol.

- ✗ Pat is 2nd.: Sol and Rae can be 2nd and 3rd
- ✗ Rae is 4th.: Rae can be 3rd
- ✗ Sol is right behind Pat.: Sol can be in front of Pat

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 120 orders checked. Automatic: the check function accepts only the keyed choice.

S7-F1-Q4 · question page 151

LOGIC.ORDER.01 · Orderings with "next to" and "not next to"

Four friends sit in seats 1 to 4 in a row. Ari is not next to Bo. Cy sits in seat 2. Bo is in a higher seat number than Cy. Di is not in seat 4. Give each friend's seat.

Answer: Ari: 1; Bo: 4; Cy: 2; Di: 3

Reasoning: Cy is in seat 2. Bo is in seat 3 or 4. If Bo were in 3, Ari (not next to Bo) would need seat 1, leaving seat 4 for Di, which is not allowed. So Bo is in 4; Ari is not in 3, so Ari is in 1 and Di in 3.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S7-F1-Q5 · question page 152

LOGIC.ORDER.01 · Orderings with "next to" and "not next to"

If five people stand in a line and three of them must stand next to each other in a fixed order, there are exactly 6 possible lines.

Answer: True

Reasoning: Treat the three as one block: 3 units (the block and two people) can be ordered in 3 x 2 x 1 = 6 ways.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S7-F2-Q1 · question page 153

LOGIC.ORDER.01 · Count and compare orderings

Five students, P, Q, R, S and T, stand in a line. P is somewhere in front of Q, and R is somewhere in front of S. How many different lines are possible?

Answer: 30

Reasoning: There are 120 lines without clues. Half have P before Q, and of those half have R before S: 120 / 4 = 30.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S7-F2-Q2 · question page 154

LOGIC.ORDER.01 · Count and compare orderings

Six relay swimmers, A to F, swim in order. B swims right after E. D swims before A. C swims last. F swims first. A swims right after B. Give the swimming order.

Answer: F → D → E → B → A → C

Reasoning: F is first and C last. E, B, A form a block (B right after E, A right after B) in positions 2-4 or 3-5. If the block were in 2-4, D would be 5th, after A, which is not allowed. So the block is in 3-5 and D is 2nd: F, D, E, B, A, C.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 720 orderings of 6 items enumerated; 1 valid.

S7-F2-Q3 · question page 155

LOGIC.ORDER.01 · Count and compare orderings

Four people, W, X, Y and Z, stand in a line. Clues: (1) W is not first. (2) X is right after W. (3) Y is last. Which clue could be removed and still leave exactly one possible line?

Answer: A. No clue can be removed.

Reasoning: With all three clues the only line is Z, W, X, Y. Removing any one clue leaves at least two lines, so every clue is needed.

- ✗ Clue 1: without it, both W, X, Z, Y and Z, W, X, Y fit
- ✗ Clue 2: without it, four lines fit (for example X, W, Z, Y)
- ✗ Clue 3: without it, four lines fit (for example Z, W, X, Y and Y, W, X, Z)

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: each clue removed in turn and all 24 lines recounted. Automatic: the check function accepts only the keyed choice.

S7-F2-Q4 · question page 156

LOGIC.ORDER.01 · Count and compare orderings

Seven people stand in a line, numbered 1 to 7 from the front. Ana is in position a and Ben is in position b. Ana is in front of Ben, exactly two people stand between them, and Ben is not last. Give one possible pair of positions.

Answer: any values that satisfy: b - a == 3 and b <= 6 and a >= 1 and isInteger(a) and isInteger(b) (example: a = 2, b = 5)

Reasoning: Two people between means b = a + 3. Ben not last means b ≤ 6, so (1, 4), (2, 5) or (3, 6).

Tempting answers the rule rejects: a = 4, b = 7; a = 2, b = 4; a = 5, b = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 3 valid answers.

S7-F2-Q5 · question page 157

LOGIC.ORDER.01 · Count and compare orderings

With the clues "J is before K" and "K is before L" for three people in a line, the order is fully determined.

Answer: True

Reasoning: J, K, L is the only order.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-M-Q1 · question page 159

LOGIC.CONNECTIVE.02 · Truth values of and, inclusive or, exclusive or

"7 is odd and 7 is greater than 10" is true.

Answer: False

Reasoning: An "and" statement needs both parts true; 7 > 10 is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-M-Q2 · question page 160

LOGIC.CONNECTIVE.02 · Truth values of and, inclusive or, exclusive or

"7 is odd or 7 is greater than 10" is true (with "or" meaning at least one).

Answer: True

Reasoning: An inclusive "or" needs at least one part true; 7 is odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-M-Q3 · question page 161

LOGIC.CONNECTIVE.02 · Truth values of and, inclusive or, exclusive or

A sign says: "Each meal comes with soup or salad, but not both." Which order follows the sign?

Answer: A. a meal with salad and no soup

Reasoning: This is an exclusive "or": exactly one of soup and salad.

- ✗ a meal with soup and salad: "not both" rules this out
- ✗ a meal with neither soup nor salad: the meal must come with one of them
- ✗ a meal with two salads and a soup: includes both soup and salad

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S8-M-Q4 · question page 162

LOGIC.CONNECTIVE.02 · Truth values of and, inclusive or, exclusive or

Give a whole number n that makes "n is even or n is greater than 20" true but "n is even and n is greater than 20" false.

Answer: any values that satisfy: (isEven(n) or n > 20) and not (isEven(n) and n > 20) and isInteger(n) (example: n = 4)

Reasoning: Exactly one part must be true: an even number up to 20 (like 4) or an odd number greater than 20 (like 25).

Tempting answers the rule rejects: n = 22; n = 7; n = 30

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 20 valid answers.

S8-M-Q5 · question page 163

LOGIC.CONNECTIVE.02 · Truth values of and, inclusive or, exclusive or

P is true and Q is false. Which statement is FALSE?

Answer: D. P and Q

Reasoning: "P and Q" needs both true, but Q is false.

- ✗ P or Q: one part is true, so the "or" is true
- ✗ P exclusive-or Q: exactly one part is true, so it is true
- ✗ P and not Q: P is true and not Q is true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S8-B1-Q1 · question page 164

LOGIC.CONNECTIVE.01 · Truth values of the parts

"12 is a multiple of 5" is true.

Answer: False

Reasoning: 12 = 5 x 2 + 2.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-B1-Q2 · question page 165

LOGIC.CONNECTIVE.01 · Truth values of the parts

"-3 < 1" is true.

Answer: True

Reasoning: -3 is to the left of 1.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-B1-Q3 · question page 166

LOGIC.CONNECTIVE.01 · Truth values of the parts

"A triangle has 4 sides" is true.

Answer: False

Reasoning: A triangle has 3 sides.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S8-B1-Q4 · question page 167

LOGIC.CONNECTIVE.01 · Truth values of the parts

"0 is even" is true.

Answer: True

Reasoning: 0 = 2 x 0.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-B1-Q5 · question page 168

LOGIC.CONNECTIVE.01 · Truth values of the parts

Which part of "9 is odd and 9 is prime" is false?

Answer: B. 9 is prime

Reasoning: 9 = 3 x 3.

- ✗ 9 is odd: that part is true
- ✗ both parts: 9 is odd is true
- ✗ neither part: 9 = 3 x 3, so it is not prime

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S8-B2-Q1 · question page 169

LOGIC.CONNECTIVE.02 · Inclusive or exclusive?

With inclusive "or", "P or Q" is false when both P and Q are true.

Answer: False

Reasoning: Inclusive "or" needs at least one part true; with both true it is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S8-B2-Q2 · question page 170

LOGIC.CONNECTIVE.02 · Inclusive or exclusive?

With exclusive "or", "P or Q" is true when both P and Q are true.

Answer: False

Reasoning: Exclusive means exactly one.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S8-B2-Q3 · question page 171

LOGIC.CONNECTIVE.02 · Inclusive or exclusive?

A rule says "You may bring a pencil or a pen (or both)." Which kind of "or" is this?

Answer: C. inclusive

Reasoning: "Or both" signals inclusive "or".

- ✗ exclusive: "or both" allows both
- ✗ neither kind: it is an "or"
- ✗ it cannot be told: "or both" tells us

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S8-B2-Q4 · question page 172

LOGIC.CONNECTIVE.02 · Inclusive or exclusive?

"The light is either on or off" describes two cases that can never both happen. Which "or" fits best?

Answer: B. exclusive

Reasoning: Exactly one of the two is true.

- ✗ inclusive: both cannot be true at once, so exactly one holds
- ✗ neither: it is an "or" statement

✗ "and": it is not a conjunction

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S8-B2-Q5 · question page 173

LOGIC.CONNECTIVE.02 · Inclusive or exclusive?

When P is false and Q is false, exclusive "P or Q" is true.

Answer: False

Reasoning: Exclusive "or" needs exactly one part true; with neither true it is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S8-F1-Q1 · question page 174

LOGIC.CONNECTIVE.02 · Compound statements with three parts

"(3 > 2 and 5 > 9) or 4 is even" is true.

Answer: True

Reasoning: The bracket is false (5 > 9 is false), but 4 is even, so the "or" is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-F1-Q2 · question page 175

LOGIC.CONNECTIVE.02 · Compound statements with three parts

"3 > 2 and (5 > 9 or 4 is odd)" is true.

Answer: False

Reasoning: The bracket is false (both parts false), so the "and" is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-F1-Q3 · question page 176

LOGIC.CONNECTIVE.02 · Compound statements with three parts

For how many whole numbers n from 1 to 20 is "n is a multiple of 3 exclusive-or n is a multiple of 4" true?

Answer: 9

Reasoning: Multiples of 3: 6 numbers; of 4: 5 numbers; of both (12): 1 number. Exactly one: 6 + 5 - 2 x 1 = 9.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S8-F1-Q4 · question page 177

LOGIC.CONNECTIVE.02 · Compound statements with three parts

Complete the truth table for "P exclusive-or Q" and for "P or Q" (inclusive).

Answer: F T / T T / T T / F F

Reasoning: They differ only in the first row, where both parts are true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S8-F1-Q5 · question page 178

LOGIC.CONNECTIVE.02 · Compound statements with three parts

Statement S: "P or Q" (inclusive). S is false. What can be concluded?

Answer: A. P is false and Q is false.

Reasoning: An inclusive "or" is false only when both parts are false.

- ✗ P is false or Q is false.: true, but it does not say enough: both must be false
- ✗ Exactly one of P and Q is false.: both are false
- ✗ P is true and Q is false.: then S would be true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S8-F2-Q1 · question page 179

LOGIC.CONNECTIVE.02 · Design statements with given truth values

P, Q and R are statements. In how many of the 8 combinations of truth values is "(P or Q) and not R" true?

Answer: 3

Reasoning: R must be false (4 cases), and at least one of P, Q true: 3 of those 4.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S8-F2-Q2 · question page 180

LOGIC.CONNECTIVE.02 · Design statements with given truth values

Give whole numbers a and b with "a > b exclusive-or a + b is even" true and both numbers between 1 and 9.

Answer: any values that satisfy: ((a > b) != isEven(a + b)) and a >= 1 and a <= 9 and b >= 1 and b <= 9 and isInteger(a) and isInteger(b) (example: a = 5, b = 2)

Reasoning: Exactly one of "a > b" and "a + b is even" must hold, for example a = 5, b = 2 (a > b, sum 7 odd) or a = 2, b = 4 (a < b, sum even).

Tempting answers the rule rejects: a = 5, b = 3; a = 2, b = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 45 valid answers.

S8-F2-Q3 · question page 181

LOGIC.CONNECTIVE.02 · Design statements with given truth values

"P exclusive-or Q" has the same truth value as "(P or Q) and not (P and Q)" in every row of a truth table.

Answer: True

Reasoning: Both are true exactly when one part is true and the other false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-F2-Q4 · question page 182

LOGIC.CONNECTIVE.02 · Design statements with given truth values

Three lamps A, B and C are each on or off. Exactly one of these statements is true: (1) "A is on or B is on." (2) "C is on." (3) "A is on exclusive-or C is on." Which lamps are on?

Answer: D. B only

Reasoning: Check the 8 patterns. If C is on, (2) is true, so (1) and (3) must be false: (3) false needs A on, but then (1) is true. So C is off and (2) is false; exactly one of (1), (3) is true. (3) is true exactly when A is on, and then (1) is also true. So A is off, (3) is false, and (1) must be true: B is on.

✗ A only: then (1) and (3) are both true

✗ A and B: then (1) and (3) are both true

✗ none: then all three statements are false

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 8 on/off patterns checked; only B on fits. Automatic: the check function accepts only the keyed choice.

S8-F2-Q5 · question page 183

LOGIC.CONNECTIVE.02 · Design statements with given truth values

For how many whole numbers n from 1 to 30 is "n is odd and (n is a multiple of 3 or n is a multiple of 5)" true?

Answer: 7

Reasoning: Odd multiples of 3: 3, 9, 15, 21, 27. Odd multiples of 5: 5, 15, 25. Together, counting 15 once: 3, 5, 9, 15, 21, 25, 27, which is 7 numbers.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S9-M-Q1 · question page 185

LOGIC.SPATIAL.02 · Distinguish rotations from reflections

Which figure shows the shape TURNED (rotated in the page), not flipped over? (Only the shape matters, not where it sits in its box.)

Answer: C. Figure C

Reasoning: Turning never changes which way the shape "curls". Figure C is the shape after a quarter-turn clockwise; A, B and D are mirror images, which no turn can produce for this shape.

✗ Figure A: a left-right flip (reflection)

✗ Figure B: a top-bottom flip (reflection)
✗ Figure D: a flip across a diagonal (reflection)

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S9-M-Q2 · question page 186

LOGIC.SPATIAL.02 · Distinguish rotations from reflections

Three of the figures are the shape turned (rotated). One is the shape flipped over (a mirror image). Which figure is the flipped one?

Answer: D. Figure D

Reasoning: Turn each candidate back to the starting position: A, B and C match the shape; D can only be matched by flipping it over.

✗ Figure A: a quarter-turn clockwise
✗ Figure B: a half-turn
✗ Figure C: a quarter-turn counterclockwise

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S9-M-Q3 · question page 187

LOGIC.SPATIAL.02 · Distinguish rotations from reflections

The shape is turned a quarter-turn CLOCKWISE (90 degrees, like a clock hand moving from 12 to 3). Which figure shows the result?

Answer: B. Figure B

Reasoning: After a clockwise quarter-turn, the left column of the shape becomes its top row, read from left to right going upward; figure B is that result.

✗ Figure A: a quarter-turn counterclockwise

✗ Figure C: a half-turn

✗ Figure D: a mirror image, not a turn

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S9-M-Q4 · question page 188

LOGIC.SPATIAL.02 · Distinguish rotations from reflections

Capital letters are turned upside down by a half-turn (180 degrees in the page). In the block capitals used here, exactly these letters look the same after a half-turn: H I N O S X Z. Which letter looks the same after the half-turn?

Answer: C. N

Reasoning: N is on the stated list; M, E and A are not.

✗ M: a half-turn of M looks like W

✗ E: a half-turn of E faces the other way and is upside down

✗ A: a half-turn of A is upside down

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S9-M-Q5 · question page 189

LOGIC.SPATIAL.02 · Distinguish rotations from reflections

A 4 by 4 grid has shaded cells at (1,1), (1,2), (1,3) and (2,1), written as row,column with row 1 at the top. The whole grid is turned a quarter-turn clockwise about its center. Mark the cells where the shaded cells end up.

Answer: 1,3, 1,4, 2,4, 3,4

Reasoning: A clockwise quarter-turn of a 4 by 4 grid sends row r, column c to row c, column 5 - r. So (1,1) -> (1,4), (1,2) -> (2,4), (1,3) -> (3,4), (2,1) -> (1,3).

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each cell mapped by (r, c) -> (c, 5 - r).

S9-B1-Q1 · question page 190

LOGIC.SPATIAL.02 · Strategy: follow one marked corner

A quarter-turn clockwise followed by another quarter-turn clockwise is the same as a half-turn.

Answer: True

Reasoning: 90 + 90 = 180 degrees.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S9-B1-Q2 · question page 191

LOGIC.SPATIAL.02 · Strategy: follow one marked corner

An arrow points up. It is turned a quarter-turn clockwise. Which way does it point now?

Answer: A. right

Reasoning: Up turns to the right, like a clock hand from 12 to 3.

- ✗ left: that is a quarter-turn counterclockwise
- ✗ down: that is a half-turn
- ✗ up: it has turned

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S9-B1-Q3 · question page 192

LOGIC.SPATIAL.02 · Strategy: follow one marked corner

An arrow points right. It is reflected in a vertical mirror line. Which way does it point now?

Answer: B. left

Reasoning: Left and right swap in a vertical mirror.

- ✗ right: a vertical mirror swaps left and right
- ✗ up: reflections do not turn it sideways
- ✗ down: reflections in a vertical line do not change up and down

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S9-B1-Q4 · question page 193

LOGIC.SPATIAL.02 · Strategy: follow one marked corner

Flipping a shape over twice in the same mirror line brings it back to where it started.

Answer: True

Reasoning: The second flip undoes the first.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S9-B1-Q5 · question page 194

LOGIC.SPATIAL.02 · Strategy: follow one marked corner

How many quarter-turns clockwise bring a shape back to its starting position for the first time, if the shape has no symmetry at all?

Answer: 4

Reasoning: Four quarter-turns make a full turn.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S9-F1-Q1 · question page 195

LOGIC.SPATIAL.02 · Combine turns and flips

Reflecting a shape in a vertical line and then in a horizontal line gives the same result as a half-turn.

Answer: True

Reasoning: (x, y) -> (-x, y) -> (-x, -y), which is a half-turn.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S9-F1-Q2 · question page 196

LOGIC.SPATIAL.02 · Combine turns and flips

A shape is turned a quarter-turn clockwise three times. Which single move gives the same result?

Answer: C. a quarter-turn counterclockwise

Reasoning: 270 degrees clockwise is the same as 90 degrees counterclockwise.

- ✗ a half-turn: three quarter-turns are 270 degrees, not 180
- ✗ a quarter-turn clockwise: that is only one of the three turns
- ✗ a reflection: turns never produce a mirror image of this shape

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S9-F1-Q3 · question page 197

LOGIC.SPATIAL.02 · Combine turns and flips

A word written in capital letters is turned a half-turn (180 degrees). In the block capitals used here, exactly these letters look the same after a half-turn: H I N O S X Z. A half-turn also

reverses the order of the letters. Which word looks exactly the same after the turn?

Answer: D. SOS

Reasoning: The word must read the same backward and use only listed letters: S, O, S.

- ✗ NOW: W is not on the list
- ✗ ZOO: turned, it reads OOZ
- ✗ MOM: M turns into W

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S9-F1-Q4 · question page 198

LOGIC.SPATIAL.02 · Combine turns and flips

A 4 by 4 grid has shaded cells at (1,2), (2,2), (3,2) and (3,3), written as row,column with row 1 at the top. The grid is given a half-turn about its center. Mark where the shaded cells end up.

Answer: 2,2, 2,3, 3,3, 4,3

Reasoning: A half-turn sends (r, c) to (5 - r, 5 - c): (4,3), (3,3), (2,3), (2,2).

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each cell mapped by (r, c) -> (5 - r, 5 - c).

S9-F1-Q5 · question page 199

LOGIC.SPATIAL.02 · Combine turns and flips

A shape has no symmetry. How many different-looking positions can it be put in using turns AND flips (on a grid, counting each distinct picture once)?

Answer: 8

Reasoning: 4 turns of the shape and 4 turns of its mirror image, all

different when the shape has no symmetry.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S9-F2-Q1 · question page 200

LOGIC.SPATIAL.02 · Symmetry under turns and flips

Of the 8 moves of a square grid figure (4 turns: 0, 90, 180, 270 degrees; and 4 flips: vertical, horizontal and both diagonals), how many leave a plus-shaped figure of 5 cells (a center cell with one cell on each side) unchanged?

Answer: 8

Reasoning: The plus shape has every symmetry of the square.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S9-F2-Q2 · question page 201

LOGIC.SPATIAL.02 · Symmetry under turns and flips

For which figure can the mirror image be produced by turning the figure (without flipping it over)? Exactly two of the four figures have this property; choose the answer that names both.

Answer: A. B and D

Reasoning: A figure's mirror image is a turn of it exactly when the figure has a mirror line itself. B (a T) and D (a staircase) each have a mirror line; A (an L) and C (a Z) do not.

- ✗ A and C: their mirror images cannot be produced by any turn
- ✗ A and B: A's mirror image cannot be produced by a turn
- ✗ C and D: C's mirror image cannot be produced by a turn

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S9-F2-Q3 · question page 202

LOGIC.SPATIAL.02 · Symmetry under turns and flips

An S-shaped figure of 4 cells (two cells in a row, and two more cells one row down, shifted one place left) is turned and flipped. Which statement is true?

Answer: C. A half-turn leaves it unchanged, but no flip does.

Reasoning: The S shape has half-turn symmetry only.

- ✗ A quarter-turn leaves it unchanged.: a quarter-turn stands it upright
- ✗ A flip in a vertical line leaves it unchanged.: that turns the S into a Z
- ✗ No move other than doing nothing leaves it unchanged.: the half-turn also does

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S9-F2-Q4 · question page 203

LOGIC.SPATIAL.02 · Symmetry under turns and flips

If a figure looks the same after a quarter-turn, then it also looks the same after a half-turn.

Answer: True

Reasoning: A half-turn is two quarter-turns; each leaves the figure unchanged.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S9-F2-Q5 · question page 204

LOGIC.SPATIAL.02 · Symmetry under turns and flips

How many of the 7 shapes made of 4 grid cells joined edge to edge (counting a shape and its turns as the same, but its mirror image as different) look the same after a half-turn?

Answer: 4

Reasoning: The 7 shapes: straight line, square, T, L, J, S, Z. Those unchanged by a half-turn: straight line, square, S and Z.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S10-M-Q1 · question page 206

LOGIC.ARGUMENT.02 · Tell deductive conclusions from inductive ones

Which argument is DEDUCTIVE (the conclusion must be true if the premises are true)?

Answer: B. Every prime number greater than 2 is odd. 31 is a prime number greater than 2. So 31 is odd.

Reasoning: Only the first conclusion is forced by its premises; the others generalize from examples or from "most".

- ✗ The last 20 buses were on time. So the next bus will be on time.: past cases make the conclusion likely, not certain
- ✗ Every swan I have seen is white. So all swans are white.: a pattern in examples (induction)
- ✗ Most students like recess. So Ana likes recess.: "most" gives probability, not certainty

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S10-M-Q2 · question page 207

LOGIC.ARGUMENT.02 · Tell deductive conclusions from inductive ones

"1 + 3 = 4, 1 + 3 + 5 = 9, 1 + 3 + 5 + 7 = 16. So the sum of the first n odd numbers is always n x n." What kind of reasoning is this?

Answer: D. inductive: it generalizes from a few examples

Reasoning: The pattern is in fact true, but the argument only checks three cases: that is inductive reasoning.

- ✗ deductive: the examples prove it for every n: checking a few cases does not prove every case
- ✗ neither: it is not an argument: it gives reasons for a conclusion
- ✗ deductive: the conclusion is true: a true conclusion does not make the reasoning deductive

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-M-Q3 · question page 208

LOGIC.ARGUMENT.02 · Tell deductive conclusions from inductive ones

In an inductive argument, the premises can all be true while the conclusion is false.

Answer: True

Reasoning: Induction makes the conclusion likely, not certain; a new case can break the pattern.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-M-Q4 · question page 209

LOGIC.ARGUMENT.02 · Tell deductive conclusions from inductive ones

n x n + n + 11 is prime for n = 1, 2, 3, ..., 9. A student concludes it is prime for every whole number n. Which value of n shows the conclusion is false?

Answer: A. 10

Reasoning: 10 x 10 + 10 + 11 = 121 = 11 x 11, which is not prime, so the conclusion is false.

- ✗ 8: 64 + 8 + 11 = 83 is prime
- ✗ 4: 16 + 4 + 11 = 31 is prime
- ✗ 9: 81 + 9 + 11 = 101 is prime

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S10-M-Q5 · question page 210

LOGIC.ARGUMENT.02 · Tell deductive conclusions from inductive ones

Which conclusion follows DEDUCTIVELY from "All members of the chess club are in grade 10 or grade 11, and Jun is a member of the chess club"?

Answer: B. Jun is in grade 10 or grade 11.

Reasoning: Only this conclusion is forced by the premises.

- ✗ Jun is in grade 11.: he could be in grade 10
- ✗ Jun is good at chess.: not stated
- ✗ Most grade 10 students are in the chess club.: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-B1-Q1 · question page 211

LOGIC.ARGUMENT.01 · Premises and conclusions

"All the cookies are gone, so someone ate them." Which part is the conclusion?

Answer: D. someone ate them

Reasoning: "So" introduces the conclusion.

- ✗ all the cookies are gone: the reason given
- ✗ cookies are tasty: not stated
- ✗ nobody ate them: the opposite of the conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-B1-Q2 · question page 212

LOGIC.ARGUMENT.01 · Premises and conclusions

A premise is the statement that an argument is trying to prove.

Answer: False

Reasoning: The statement an argument tries to prove is its conclusion; premises are the reasons offered for it.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-B1-Q3 · question page 213

LOGIC.ARGUMENT.01 · Premises and conclusions

"It rained every day this week. So it will rain tomorrow." What is the premise?

Answer: A. It rained every day this week.

Reasoning: The reason given first.

- ✗ It will rain tomorrow.: the conclusion
- ✗ Rain is good for plants.: not stated
- ✗ It never rains.: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-B1-Q4 · question page 214

LOGIC.ARGUMENT.01 · Premises and conclusions

Every argument has at least one premise and one conclusion.

Answer: True

Reasoning: That is what makes it an argument.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-B1-Q5 · question page 215

LOGIC.ARGUMENT.01 · Premises and conclusions

"Since 12 is even, 12 is divisible by 2." What is the conclusion?

Answer: C. 12 is divisible by 2

Reasoning: The claim supported by the reason.

- ✗ 12 is even: the reason after "since"
- ✗ 12 is odd: false and not stated
- ✗ 2 is even: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-B2-Q1 · question page 216

LOGIC.ARGUMENT.02 · Certain or only likely?

"Most tickets in the box are red, and I drew a ticket from the box" makes "my ticket is red" certain.

Answer: False

Reasoning: If only most tickets are red, the drawn ticket could be one of the others, so "my ticket is red" is likely but not certain.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-B2-Q2 · question page 217

LOGIC.ARGUMENT.02 · Certain or only likely?

"Most tickets in the box are red, and I drew a ticket" makes "my ticket is red" certain.

Answer: False

Reasoning: It is likely, not certain.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-B2-Q3 · question page 218

LOGIC.ARGUMENT.02 · Certain or only likely?

Which word signals that a conclusion is only likely?

Answer: B. probably

Reasoning: "Probably" marks a likely conclusion.

- ✗ must: signals certainty
- ✗ therefore: signals a conclusion, not its strength
- ✗ every: a premise word

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S10-B2-Q4 · question page 219

LOGIC.ARGUMENT.02 · Certain or only likely?

If a pattern holds for 100 examples, it must hold for the 101st.

Answer: False

Reasoning: Examples alone never guarantee the next case.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-B2-Q5 · question page 220

LOGIC.ARGUMENT.02 · Certain or only likely?

A die has been rolled 5 times and showed an even number each time. Which conclusion is certain?

Answer: C. Each of those 5 rolls was even.

Reasoning: Only the statement about what already happened is certain.

- ✗ The next roll will be even.: not certain
- ✗ The die only has even numbers.: not certain
- ✗ The die is unfair.: not certain

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-F1-Q1 · question page 221

LOGIC.ARGUMENT.02 · Patterns that break

The claim "n x n - n + 17 is prime for every whole number n" holds for n = 1 to 16. Find a whole number n that makes n x n - n + 17 NOT prime.

Answer: any values that satisfy: not isPrime(n*n - n + 17) and n >= 1 and isInteger(n) (example: n = 17)

Reasoning: For n = 17: 17 x 17 - 17 + 17 = 289 = 17 x 17. Other values also work (for example 18: 323 = 17 x 19).

Tempting answers the rule rejects: n = 1; n = 5; n = 16

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 8 valid answers.

S10-F1-Q2 · question page 222

LOGIC.ARGUMENT.02 · Patterns that break

Points are placed on a circle and every pair is joined by a straight line. With 1, 2, 3, 4, 5 points (in general position), the circle is cut into 1, 2, 4, 8, 16 regions. A student predicts 32 regions for 6 points. What is the actual number?

Answer: A. 31

Reasoning: The region count is 1 + C(n, 2) + C(n, 4): for n = 6, 1 + 15 + 15 = 31. The early pattern of doubling was only inductive evidence.

- ✗ 32: the doubling pattern breaks at 6 points
- ✗ 30: one too few
- ✗ 36: not the count

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S10-F1-Q3 · question page 223

LOGIC.ARGUMENT.02 · Patterns that break

A deductive argument with true premises can have a false conclusion.

Answer: False

Reasoning: In a valid deductive argument, true premises guarantee a

true conclusion.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-F1-Q4 · question page 224

LOGIC.ARGUMENT.02 · Patterns that break

Which statement is the strongest inductive argument?

Answer: D. All 500 randomly chosen bolts from the batch passed, so nearly all bolts in the batch pass.

Reasoning: A large random sample gives the strongest support for a general claim about the batch.

- ✗ My two friends like the song, so everyone does.: two hand-picked examples
- ✗ The first bolt passed, so all bolts pass.: one example
- ✗ All bolts in the batch passed, so this bolt passed.: that is deductive, not inductive

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-F1-Q5 · question page 225

LOGIC.ARGUMENT.02 · Patterns that break

Which reasoning is deductive?

Answer: B. All squares are rectangles; this figure is a square; so it is a rectangle.

Reasoning: Only the first is forced by its premises.

- ✗ This figure looks like a rectangle, so it probably is one.: based on

appearance

✗ Every rectangle I measured had equal diagonals, so all rectangles do.: from measured examples

✗ Most rectangles in the book are squares, so this one is.: from "most"

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-F2-Q1 · question page 226

LOGIC.ARGUMENT.02 · From pattern to proof

The expression n x n + n + 41 is prime for n = 0, 1, 2, ..., 39. What is the smallest whole number n for which it is NOT prime?

Answer: 40

Reasoning: n = 40 gives 1681 = 41 x 41.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S10-F2-Q2 · question page 227

LOGIC.ARGUMENT.02 · From pattern to proof

"For every whole number n, n x n - n is even, because 2 x 2 - 2 = 2, 3 x 3 - 3 = 6 and 4 x 4 - 4 = 12 are even." Which statement about this argument is correct?

Answer: A. The claim is true, but three examples do not prove it.

Reasoning: n x n - n = n(n - 1) is a product of two consecutive whole numbers, one of them even, so the claim is true; but the argument only checks examples, which does not prove it for every n.

✗ The argument proves the claim, because three cases are checked.: cases do not prove every n

✗ The claim is false, because 1 x 1 - 1 = 0.: 0 is even

✗ The claim is false for large n.: n(n - 1) is always even

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-F2-Q3 · question page 228

LOGIC.ARGUMENT.02 · From pattern to proof

If every case of a finite list has been checked, the conclusion about that list is deductive.

Answer: True

Reasoning: Checking ALL cases of a finite list is a complete proof (proof by cases).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-F2-Q4 · question page 229

LOGIC.ARGUMENT.02 · From pattern to proof

Give a whole number n from 1 to 20 for which "2 to the power n, minus 1, is prime" is FALSE, even though n is prime. (2 to the power n is 2 multiplied by itself n times.)

Answer: any values that satisfy: isPrime(n) and not isPrime(2^n - 1) and n <= 20 (example: n = 11)

Reasoning: 2 to the 11th minus 1 = 2047 = 23 x 89. The pattern "n prime gives 2^n - 1 prime" holds for 2, 3, 5, 7 but fails at 11, the only such n up to 20.

Tempting answers the rule rejects: n = 7; n = 5; n = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S10-F2-Q5 · question page 230

LOGIC.ARGUMENT.02 · From pattern to proof

A student checks that every whole number from 4 to 100 that is even can be written as a sum of two primes. Which conclusion is justified?

Answer: C. Every even number from 4 to 100 is a sum of two primes.

Reasoning: Only the statement about the checked range is established.

- ✗ Every even number greater than 2 is a sum of two primes.: checking up to 100 does not cover all even numbers
- ✗ No odd number is a sum of two primes.: 2 + 3 = 5
- ✗ Every even number is a sum of two primes.: 2 is even and is not a sum of two primes

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S11-M-Q1 · question page 232

LOGIC.WORD.01 · Use letter counts, positions and order to test word patterns

Which word has its letters in alphabetical order (each letter the same as or later in the alphabet than the one before)?

Answer: D. ALMOST

Reasoning: A, L, M, O, S, T are in alphabetical order.

- ✗ BEGAN: A comes after G
- ✗ FLOWER: E comes after W in the word but before W in the alphabet
- ✗ CHART: A comes after H

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-M-Q2 · question page 233

LOGIC.WORD.01 · Use letter counts, positions and order to test word patterns

How many times does the letter S appear in the word ASSESSMENTS?

Answer: 5

Reasoning: A S S E S S M E N T S: positions 2, 3, 5, 6 and 11.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-M-Q3 · question page 234

LOGIC.WORD.01 · Use letter counts, positions and order to test word patterns

Number words are written in capital letters (ONE, TWO, ..., TEN). Which number word has exactly as many letters as the number it names?

Answer: A. FOUR

Reasoning: F-O-U-R has 4 letters.

- ✗ FIVE: 4 letters, value 5
- ✗ THREE: 5 letters, value 3
- ✗ TWO: 3 letters, value 2

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-M-Q4 · question page 235

LOGIC.WORD.01 · Use letter counts, positions and order to test word patterns

Take the first letter of each word: "Every Apple Tastes Sweet." What word do the first letters spell?

Answer: C. EATS

Reasoning: E, A, T, S in order: EATS.

- ✗ EAST: the order is E, A, T, S
- ✗ SEAT: the letters must stay in order
- ✗ TEAS: the letters must stay in order

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-M-Q5 · question page 236

LOGIC.WORD.01 · Use letter counts, positions and order to test word patterns

Write the letters of STEAM in alphabetical order. Which letter is in the middle position?

Answer: B. M

Reasoning: A, E, M, S, T: the 3rd letter is M.

- ✗ E: E is 2nd
- ✗ A: A is 1st
- ✗ S: S is 4th

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-B1-Q1 · question page 237

LOGIC.WORD.01 · Strategy: list the letters with their positions

The letters of BELLOW are in alphabetical order.

Answer: True

Reasoning: B, E, L, L, O, W: each the same as or later than the one

before.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S11-B1-Q2 · question page 238

LOGIC.WORD.01 · Strategy: list the letters with their positions

How many times does the letter E appear in REFERENCE?

Answer: 4

Reasoning: R E F E R E N C E.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-B1-Q3 · question page 239

LOGIC.WORD.01 · Strategy: list the letters with their positions

What is the 4th letter of the word PLANET?

Answer: D. N

Reasoning: P L A N.

- ✗ A: the 3rd letter
- ✗ E: the 5th letter
- ✗ L: the 2nd letter

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-B1-Q4 · question page 240

LOGIC.WORD.01 · Strategy: list the letters with their positions

Take the first letter of each word: "Owls Watch Lakes." What do the first letters spell?

Answer: B. OWL

Reasoning: O, W, L.

- ✗ LOW: the order changed
- ✗ OLW: the letters are out of order
- ✗ WOL: the order changed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S11-B1-Q5 · question page 241

LOGIC.WORD.01 · Strategy: list the letters with their positions

How many letters does the word EIGHT have?

Answer: 5

Reasoning: E I G H T.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F1-Q1 · question page 242

LOGIC.WORD.01 · Patterns across several words

In the list GO, CAT, LAMP, PLANT, each word has one more letter than the word before. Which word could come next?

Answer: C. GARDEN

Reasoning: GO has 2 letters, CAT 3, LAMP 4, PLANT 5, so the next word needs 6 letters.

✗ HOUSE: 5 letters
✗ BALLOON: 7 letters
✗ TREE: 4 letters

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-F1-Q2 · question page 243

LOGIC.WORD.01 · Patterns across several words

How many of the number words ONE, TWO, THREE, ..., TEN have their letters in alphabetical order?

Answer: 0

Reasoning: Check each: none of them is in alphabetical order (for example ONE: N comes before O).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F1-Q3 · question page 244

LOGIC.WORD.01 · Patterns across several words

Which word has its letters in strictly reverse alphabetical order (each letter earlier in the alphabet than the one before it)?

Answer: D. SPOKE

Reasoning: S, P, O, K, E: each letter comes earlier in the alphabet than the one before it.

✗ STONE: T comes after S
✗ TRACK: C comes after A
✗ SPELL: L comes after E

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-F1-Q4 · question page 245

LOGIC.WORD.01 · Patterns across several words

Write the letters of each number word ONE to TEN in alphabetical order; for example, ONE becomes ENO. Which number word becomes EENSV? Give its value as a number.

Answer: 7

Reasoning: SEVEN sorted: E, E, N, S, V. No other number word has these letters.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F1-Q5 · question page 246

LOGIC.WORD.01 · Patterns across several words

Every word whose letters are in alphabetical order reads differently backward unless all its letters are the same.

Answer: True

Reasoning: Backward it is in reverse order; the two orders agree only if no two letters differ.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S11-F2-Q1 · question page 247

LOGIC.WORD.01 · Count and construct letter strings under rules

How many 3-letter strings can be made from the letters A, B, C, D (letters may repeat)

whose letters are in alphabetical order (each letter the same as or later than the one before)?

Answer: 20

Reasoning: Choosing a multiset of 3 letters from 4 kinds: C(6, 3) = 20.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F2-Q2 · question page 248

LOGIC.WORD.01 · Count and construct letter strings under rules

How many 4-letter strings using only the letters X and Y have no two equal letters next to each other?

Answer: 2

Reasoning: They must alternate: XYXY or YXYX.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F2-Q3 · question page 249

LOGIC.WORD.01 · Count and construct letter strings under rules

How many 4-letter strings using the letters A, B, C (letters may repeat) have no letter next to itself?

Answer: 24

Reasoning: 3 choices for the first letter, then 2 for each of the next three: 3 x 2 x 2 x 2 = 24.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F2-Q4 · question page 250

LOGIC.WORD.01 · Count and construct letter strings under rules

A 6-letter string uses each of the letters A, B, C exactly twice, and no letter is next to itself. Which string follows the rule?

Answer: A. ABCABC

Reasoning: Check neighbors: A-B, B-C, C-A, A-B, B-C are all different.

- ✗ AABBCC: equal letters are next to each other
- ✗ ABBCAC: B is next to B
- ✗ ACBBCA: B is next to B

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-F2-Q5 · question page 251

LOGIC.WORD.01 · Count and construct letter strings under rules

The five letters of LEVEL (L, E, V, E, L) are rearranged into 5-letter strings. How many different strings have no two equal letters next to each other?

Answer: 12

Reasoning: Place V first. V at either end: the other four positions are in a row and must alternate (LELE or ELEL), 2 ways each, so 4. V in the middle: positions 1-2 and 4-5 each need one L and one E, 2 x 2 = 4 ways. V 2nd or 4th: the single end letter is L or E and the three positions on the other side must alternate starting with the other letter, 2 ways each, so 4. Total 4 + 4 + 4 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S12-M-Q1 · question page 253

LOGIC.CONDITIONAL.01 · Find when a conditional is true and the one case that makes it false

A rule: "If a package weighs more than 5 kg, then it has a yellow label." Which package BREAKS the rule?

Answer: B. a 7 kg package with a blue label

Reasoning: A conditional is false only when the "if" part is true and the "then" part is false: heavy and not yellow.

- ✗ a 3 kg package with a yellow label: the rule says nothing about light packages
- ✗ a 3 kg package with a blue label: light packages may have any label
- ✗ a 7 kg package with a yellow label: this follows the rule

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-M-Q2 · question page 254

LOGIC.CONDITIONAL.01 · Find when a conditional is true and the one case that makes it false

Complete the truth table for "if P then Q".

Answer: T / F / T / T

Reasoning: It is false only in the row P true, Q false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S12-M-Q3 · question page 255

LOGIC.CONDITIONAL.01 · Find when a conditional is true and the one case that makes it false

Find a whole number n that is a counterexample to "If n is odd, then n is prime." (A counterexample makes the "if" part true and the "then" part false.)

Answer: any values that satisfy: isOdd(n) and not isPrime(n) and n >= 1 (example: n = 9)

Reasoning: Any odd number that is not prime works: 1, 9, 15, 21, 25, ...

Tempting answers the rule rejects: n = 7; n = 4; n = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S12-M-Q4 · question page 256

LOGIC.CONDITIONAL.01 · Find when a conditional is true and the one case that makes it false

"If 5 > 8, then 2 + 2 = 5" is true.

Answer: True

Reasoning: The "if" part is false, so the conditional is true (it makes no promise when the "if" part is false).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S12-M-Q5 · question page 257

LOGIC.CONDITIONAL.01 · Find when a conditional is true and the one case that makes it false

"If it is a weekend day, the library opens at noon." Today the library opened at 9 a.m. What follows?

Answer: C. Today is not a weekend day.

Reasoning: If it were a weekend day, the library would have opened at noon. It did not, so it is not a weekend day.

- ✗ Today is a weekend day.: then it would have opened at noon
- ✗ The rule was broken.: the rule allows 9 a.m. on weekdays
- ✗ Nothing follows.: the contrapositive gives a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B1-Q1 · question page 258

LOGIC.CONDITIONAL.01 · Find the "if" part and the "then" part

In "If the sun sets, the street lights turn on," what is the "if" part (the hypothesis)?

Answer: D. the sun sets

Reasoning: The part after "if".

- ✗ the street lights turn on: that is the "then" part
- ✗ the lights are bright: not stated
- ✗ it is night: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B1-Q2 · question page 259

LOGIC.CONDITIONAL.01 · Find the "if" part and the "then" part

In "Students may enter the lab only if they wear goggles," which is the "then" part when it is written as "If ..., then ..."?

Answer: A. they wear goggles

Reasoning: "A only if B" means "if A, then B": if a student enters, the student wears goggles.

✗ students enter the lab: that becomes the "if" part

✗ the lab is open: not stated

✗ they leave the lab: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B1-Q3 · question page 260

LOGIC.CONDITIONAL.01 · Find the "if" part and the "then" part

Rewrite "Every multiple of 10 ends in 0" as a conditional.

Answer: C. If a number is a multiple of 10, then it ends in 0.

Reasoning: "Every A is B" means "if something is A, then it is B".

✗ If a number ends in 0, then it is a multiple of 10.: the parts are swapped

✗ If a number is not a multiple of 10, then it does not end in 0.: that changes the meaning

✗ A number ends in 0 and is a multiple of 10.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B1-Q4 · question page 261

LOGIC.CONDITIONAL.01 · Find the "if" part and the "then" part

In "If x = 3, then x + 1 = 4," the hypothesis is "x + 1 = 4".

Answer: False

Reasoning: The hypothesis is "x = 3".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S12-B1-Q5 · question page 262

LOGIC.CONDITIONAL.01 · Find the "if" part and the "then" part

In "The alarm rings whenever the door opens," which part is the hypothesis?

Answer: A. the door opens

Reasoning: "Whenever the door opens" = "if the door opens".

- ✗ the alarm rings: the conclusion
- ✗ the alarm is loud: not stated
- ✗ the door closes: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B2-Q1 · question page 263

LOGIC.CONNECTIVE.01 · Truth values of the parts

"6 is a factor of 30" is true.

Answer: True

Reasoning: 30 = 6 x 5.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S12-B2-Q2 · question page 264

LOGIC.CONNECTIVE.01 · Truth values of the parts

"4 squared is 8" is true.

Answer: False

Reasoning: 4 x 4 = 16.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S12-B2-Q3 · question page 265

LOGIC.CONNECTIVE.01 · Truth values of the parts

"Every square is a triangle" is true.

Answer: False

Reasoning: A square has 4 sides.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S12-B2-Q4 · question page 266

LOGIC.CONNECTIVE.01 · Truth values of the parts

Which statement is true?

Answer: B. -5 is less than 2.

Reasoning: On a number line -5 is left of 2.

- ✗ 2 is less than -5.: 2 is to the right of -5 on the number line, so it is greater
- ✗ -5 is greater than 2.: -5 is to the left of 2 on the number line, so it is less
- ✗ -5 equals 5.: -5 and 5 are on opposite sides of 0

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B2-Q5 · question page 267

LOGIC.CONNECTIVE.01 · Truth values of the parts

"1 is a prime number" is true.

Answer: False

Reasoning: A prime has exactly two different factors; 1 has one.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S12-F1-Q1 · question page 268

LOGIC.CONDITIONAL.01 · Conditionals in mathematics

Find a real number x that makes "If x squared is greater than 9, then x is greater than 3" FALSE.

Answer: any values that satisfy: x^2 > 9 and not (x > 3) (example: x = -4)

Reasoning: The conditional is false when x squared > 9 but x ≤ 3: any x < -3 works, for example -4.

Tempting answers the rule rejects: x = 4; x = 2; x = -3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S12-F1-Q2 · question page 269

LOGIC.CONDITIONAL.01 · Conditionals in mathematics

"If a whole number is divisible by 6, then it is divisible by 3" is true for every whole number.

Answer: True

Reasoning: 6 = 2 x 3, so a multiple of 6 is a multiple of 3. No case has the "if" true and the "then" false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S12-F1-Q3 · question page 270

LOGIC.CONDITIONAL.01 · Conditionals in mathematics

For how many whole numbers n from 1 to 12 is "If n is even, then n is a multiple of 4" FALSE?

Answer: 3

Reasoning: False when n is even but not a multiple of 4: 2, 6, 10.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S12-F1-Q4 · question page 271

LOGIC.CONDITIONAL.01 · Conditionals in mathematics

P is false. Which statement MUST be true?

Answer: D. If P then Q

Reasoning: A conditional with a false "if" part is always true.

- ✗ If Q then P: false when Q is true
- ✗ P and Q: false because P is false
- ✗ Q: not determined

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S12-F1-Q5 · question page 272

LOGIC.CONDITIONAL.01 · Conditionals in mathematics

"If you score above 90, you earn a badge." Lin earned a badge. What follows?

Answer: C. Nothing about Lin's score follows.

Reasoning: The rule only says what happens above 90; badges may be earned in other ways.

- ✗ Lin scored above 90.: the rule does not say badges come only from scores above 90
- ✗ Lin scored 90 or below.: not supported
- ✗ The rule was broken.: earning a badge breaks nothing

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-F2-Q1 · question page 273

LOGIC.CONDITIONAL.01 · Nested and chained conditionals

Complete the truth table for "if P then (if Q then R)".

Answer: T T / F F / T T / T T / T T / F T / T T / T T

Reasoning: It is false only when P and Q are true and R is false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (8 rows).

S12-F2-Q2 · question page 274

LOGIC.CONDITIONAL.01 · Nested and chained conditionals

In how many of the 8 rows of a truth table for P, Q, R is "(if P then Q) and (if Q then R)" true?

Answer: 4

Reasoning: Check all 8 rows; it is true in TTT, FTT, FFT, FFF.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S12-F2-Q3 · question page 275

LOGIC.CONDITIONAL.01 · Nested and chained conditionals

"If P then (if Q then R)" has the same truth value as "if (P and Q) then R" in every row.

Answer: True

Reasoning: Both are false exactly when P and Q are true and R is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S12-F2-Q4 · question page 276

LOGIC.CONDITIONAL.01 · Nested and chained conditionals

"If P then (if Q then R)" has the same truth value as "if (if P then Q) then R" in every row.

Answer: False

Reasoning: Row P false, Q false, R false: the first is true, but "if P then Q" is true and R false, so the second is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S12-F2-Q5 · question page 277

LOGIC.CONDITIONAL.01 · Nested and chained conditionals

Find whole numbers a and b that make the statement "If a + b is even, then a and b are both even" FALSE.

Answer: any values that satisfy: isEven(a + b) and not (isEven(a) and isEven(b)) and isInteger(a) and isInteger(b) (example: a = 3, b = 5)

Reasoning: Two odd numbers have an even sum: for example 3 + 5 = 8.

Tempting answers the rule rejects: a = 2, b = 4; a = 2, b = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 25 valid answers.

S13-M-Q1 · question page 279

LOGIC.CONDITIONAL.02 · Translate "only if", "unless", "necessary" and "sufficient"

"Being at least 16 years old is NECESSARY for getting a driving permit in this town." Which conditional says the same thing?

Answer: A. If a person has a driving permit, then the person is at least 16.

Reasoning: A necessary condition is required: whoever has the permit must meet it. "B is necessary for A" means "if A, then B".

- ✗ If a person is at least 16, then the person has a driving permit.: this makes age sufficient, which was not claimed
- ✗ If a person is under 16, then the person may still get a permit.: this contradicts the statement
- ✗ A person is at least 16 and has a permit.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-M-Q2 · question page 280

LOGIC.CONDITIONAL.02 · Translate "only if", "unless", "necessary" and "sufficient"

"You may enter the museum only if you have a ticket." Which statement means the same?

Answer: D. If you enter the museum, then you have a ticket.

Reasoning: "A only if B" means "if A, then B".

- ✗ If you have a ticket, then you may enter the museum.: "only if" does not say a ticket is enough

✗ If you have no ticket, then you may still enter.: that contradicts the rule

✗ You have a ticket and you enter.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-M-Q3 · question page 281

LOGIC.CONDITIONAL.02 · Translate "only if", "unless", "necessary" and "sufficient"

In logic, "P unless Q" means "if not Q, then P". Which statement means the same as "The picnic is held unless it rains"?

Answer: B. If it does not rain, the picnic is held.

Reasoning: With the stated meaning: if it does not rain, then the picnic is held.

✗ If it rains, the picnic is held.: swaps the condition

✗ If the picnic is held, it does not rain.: that is the converse of the stated meaning

✗ If it does not rain, the picnic is not held.: adds a negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-M-Q4 · question page 282

LOGIC.CONDITIONAL.02 · Translate "only if", "unless", "necessary" and "sufficient"

Being a square is a NECESSARY condition for being a rectangle.

Answer: False

Reasoning: A 2 by 5 rectangle is not a square, so a shape can be a rectangle without being a square. Being a square is sufficient, not necessary.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-M-Q5 · question page 283

LOGIC.CONDITIONAL.02 · Translate "only if", "unless", "necessary" and "sufficient"

n is a whole number. Which condition is NECESSARY for "n is divisible by 6"?

Answer: C. n is even

Reasoning: Every multiple of 6 is even, so being even is required (necessary). It is not sufficient: 4 is even.

- ✗ n is divisible by 12: sufficient but not necessary: 6 is divisible by 6 but not by 12
- ✗ n is divisible by 9: 12 is divisible by 6 but not by 9
- ✗ n ends in the digit 6: 12 is divisible by 6 but ends in 2

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: each condition tested against every n from 1 to 200. Automatic: the check function accepts only the keyed choice.

S13-B1-Q1 · question page 284

LOGIC.CONDITIONAL.01 · Write a statement as "if ..., then ..."

Rewrite "All birds have feathers" as a conditional.

Answer: A. If an animal is a bird, then it has feathers.

Reasoning: "All A are B" means "if A, then B".

- ✗ If an animal has feathers, then it is a bird.: parts swapped
- ✗ If an animal is not a bird, then it has no feathers.: changes the meaning
- ✗ Birds and feathers.: not a statement

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-B1-Q2 · question page 285

LOGIC.CONDITIONAL.01 · Write a statement as "if ..., then ..."

Rewrite "When the bell rings, class ends" as a conditional.

Answer: D. If the bell rings, then class ends.

Reasoning: "When" introduces the condition.

- ✗ If class ends, then the bell rings.: parts swapped
- ✗ The bell rings and class ends.: not a conditional
- ✗ If the bell does not ring, class does not end.: changes the meaning

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-B1-Q3 · question page 286

LOGIC.CONDITIONAL.01 · Write a statement as "if ..., then ..."

"If x = 2, then x + x = 4" has hypothesis "x + x = 4".

Answer: False

Reasoning: The hypothesis is the part after "if", which is "x = 2". "x + x = 4" is the conclusion.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-B1-Q4 · question page 287

LOGIC.CONDITIONAL.01 · Write a statement as "if ..., then ..."

What is the conclusion of "If a number ends in 5, it is divisible by 5"?

Answer: B. it is divisible by 5

Reasoning: The "then" part.

- ✗ a number ends in 5: the hypothesis
- ✗ it is odd: not stated
- ✗ it is 5: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-B1-Q5 · question page 288

LOGIC.CONDITIONAL.01 · Write a statement as "if ..., then ..."

"Every even number is divisible by 2" can be written "If a number is even, then it is divisible by 2".

Answer: True

Reasoning: "Every A is B" = "if A, then B".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-B2-Q1 · question page 289

LOGIC.CONDITIONAL.02 · Necessary means required; sufficient means enough

Having fuel is the only thing a gas car needs in order to run.

Answer: False

Reasoning: A car with fuel can still fail to run, for example with a dead battery. Fuel is necessary, not sufficient.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-B2-Q2 · question page 290

LOGIC.CONDITIONAL.02 · Necessary means required; sufficient means enough

Having fuel is sufficient for a gas car to run.

Answer: False

Reasoning: A car with fuel may still not run (for example, a dead battery).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-B2-Q3 · question page 291

LOGIC.CONDITIONAL.02 · Necessary means required; sufficient means enough

Which is SUFFICIENT for a whole number to be even?

Answer: B. It is a multiple of 10.

Reasoning: Every multiple of 10 is even.

- ✗ It is greater than 2.: 3 is greater than 2 and odd
- ✗ It is a multiple of 3.: 3 is odd
- ✗ It ends in 1.: such numbers are odd

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S13-B2-Q4 · question page 292

LOGIC.CONDITIONAL.02 · Necessary means required; sufficient means enough

If A is sufficient for B, then whenever B happens, A happens.

Answer: False

Reasoning: Sufficient means A guarantees B. B can still happen without A.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-B2-Q5 · question page 293

LOGIC.CONDITIONAL.02 · Necessary means required; sufficient means enough

Being a mammal is ___ for being a dog.

Answer: A. necessary

Reasoning: Every dog is a mammal (required), but not every mammal is a dog.

- ✗ sufficient: a cat is a mammal but not a dog
- ✗ neither necessary nor sufficient: every dog is a mammal
- ✗ both necessary and sufficient: not sufficient

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-F1-Q1 · question page 294

LOGIC.CONDITIONAL.02 · Combine conditions

"n is even" is necessary for "n is divisible by 6". Give a whole number n that shows it is NOT sufficient: n is even but not divisible by 6.

Answer: any values that satisfy: isEven(n) and n % 6 != 0 and isInteger(n) (example: n = 4)

Reasoning: Any even number that is not a multiple of 6: 2, 4, 8, 10, ...

Tempting answers the rule rejects: n = 6; n = 12; n = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S13-F1-Q2 · question page 295

LOGIC.CONDITIONAL.02 · Combine conditions

n is a whole number. Which condition is SUFFICIENT for "n is divisible by 4"?

Answer: D. n is divisible by 8

Reasoning: Every multiple of 8 is a multiple of 4.

- ✗ n is even: 2 is even but not divisible by 4
- ✗ n is divisible by 2 and by 3: 6 is not divisible by 4
- ✗ n is greater than 4: 5 is not divisible by 4

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: tested for every n from 1 to 200. Automatic: the check function accepts only the keyed choice.

S13-F1-Q3 · question page 296

LOGIC.CONDITIONAL.02 · Combine conditions

"A triangle is equilateral only if it is isosceles (has at least two equal sides)." Which statement is correct?

Answer: C. Being isosceles is necessary for being equilateral.

Reasoning: "A only if B": B is necessary for A.

- ✗ Being isosceles is sufficient for being equilateral.: a 5-5-8 triangle is isosceles, not equilateral
- ✗ Being equilateral is necessary for being isosceles.: reverses the relation
- ✗ The two conditions mean the same thing.: they do not

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S13-F1-Q4 · question page 297

LOGIC.CONDITIONAL.02 · Combine conditions

"You get the discount unless you pay late." Using "P unless Q" = "if not Q, then P", which person is guaranteed the discount?

Answer: D. someone who did not pay late

Reasoning: If you did not pay late, you get the discount.

- ✗ someone who paid late: the statement promises nothing to late payers
- ✗ everyone: late payers are not covered
- ✗ no one: on-time payers are covered

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-F1-Q5 · question page 298

LOGIC.CONDITIONAL.02 · Combine conditions

A condition can be both necessary and sufficient for another.

Answer: True

Reasoning: For example, for a whole number, "ends in 0 or 5" is necessary and sufficient for "divisible by 5".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-F2-Q1 · question page 299

LOGIC.CONDITIONAL.02 · Chains of conditions

Facts: A is sufficient for B. B is sufficient for C. Which statement must be true?

Answer: A. C is necessary for A.

Reasoning: A -> B -> C, so A -> C: C is necessary for A (and A is sufficient for C).

- ✗ A is necessary for C.: C can hold without A
- ✗ B is sufficient for A.: reverses the first fact
- ✗ C is sufficient for A.: reverses the chain

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 8 truth assignments of A, B, C checked. Automatic: the check function accepts only the keyed choice.

S13-F2-Q2 · question page 300

LOGIC.CONDITIONAL.02 · Chains of conditions

For whole numbers n from 1 to 60, condition X: "n is divisible by 4"; condition Y: "n is divisible by 6". How many n satisfy Y but not X (showing X is not necessary for Y)?

Answer: 5

Reasoning: Multiples of 6 up to 60: 10 numbers. Those also divisible by 4 are multiples of 12: 5. So 10 - 5 = 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S13-F2-Q3 · question page 301

LOGIC.CONDITIONAL.02 · Chains of conditions

Which condition on a whole number n is BOTH necessary and sufficient for "n is divisible by 10"?

Answer: C. n is divisible by 2 and by 5

Reasoning: A number is divisible by 10 exactly when it is divisible by

both 2 and 5.

- ✗ n is divisible by 5: not sufficient: 15
- ✗ n is even: not sufficient: 4
- ✗ n is divisible by 20: not necessary: 10

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S13-F2-Q4 · question page 302

LOGIC.CONDITIONAL.02 · Chains of conditions

If A is necessary for B and B is necessary for A, then A and B are always both true or both false.

Answer: True

Reasoning: B -> A and A -> B together mean A and B have the same truth value.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-F2-Q5 · question page 303

LOGIC.CONDITIONAL.02 · Chains of conditions

Give a whole number n for which "n is a multiple of 3" holds but "n is a multiple of 9" fails, showing that being a multiple of 3 is not sufficient for being a multiple of 9.

Answer: any values that satisfy: n % 3 == 0 and n % 9 != 0 and isInteger(n) (example: n = 6)

Reasoning: Any multiple of 3 that is not a multiple of 9: 3, 6, 12, 15, ...

Tempting answers the rule rejects: n = 9; n = 18; n = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are

correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S14-M-Q1 · question page 305

LOGIC.GRID.01 · Solve a matching puzzle by combining clues

Three students, Ama, Bex and Cho, each play one instrument: drums, flute or violin (all different). Ama does not play the flute. Bex plays neither the drums nor the flute. Which instrument does each student play?

Answer: Ama: drums; Bex: violin; Cho: flute

Reasoning: Bex plays the violin (not drums, not flute). Ama does not play the flute, so Ama plays the drums, and Cho plays the flute.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S14-M-Q2 · question page 306

LOGIC.GRID.01 · Solve a matching puzzle by combining clues

Four friends, Dara, Eli, Faye and Gus, each ordered a different drink: cocoa, juice, milk or tea. Dara did not order a hot drink (cocoa and tea are hot). Eli ordered tea. Faye ordered neither cocoa nor juice. Who ordered what?

Answer: Dara: juice; Eli: tea; Faye: milk; Gus: cocoa

Reasoning: Eli: tea. Faye: not cocoa, not juice, not tea, so milk. Dara: not hot, so juice. Gus: cocoa.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S14-M-Q3 · question page 307

LOGIC.GRID.01 · Solve a matching puzzle by combining clues

Four runners wore bibs 1, 2, 3 and 4 (all different): Hal, Ivy, Jun and Kit. Ivy's number is

twice Hal's. Hal's number is even. Jun's number is odd. Kit's number is greater than Jun's. What number did each runner wear?

Answer: Hal: 2; Ivy: 4; Jun: 1; Kit: 3

Reasoning: Ivy = 2 x Hal with Hal even means Hal 2 and Ivy 4. Jun and Kit share 1 and 3; Jun is odd either way, and Kit is greater, so Jun 1 and Kit 3.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S14-M-Q4 · question page 308

LOGIC.GRID.01 · Solve a matching puzzle by combining clues

Three pets, a cat, a dog and a rabbit, belong to Lea, Max and Noa (one each). The cat's owner sits next to Max. Lea owns neither the dog nor the cat. Noa does not own the dog. Who owns which pet?

Answer: Lea: rabbit; Max: dog; Noa: cat

Reasoning: Lea owns the rabbit. Noa does not own the dog, so Noa owns the cat and Max owns the dog. (The cat's owner sits next to Max, so the cat is not Max's.)

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S14-M-Q5 · question page 309

LOGIC.GRID.01 · Solve a matching puzzle by combining clues

Four lockers, numbered 1 to 4 in a row, belong to Oz, Pia, Quin and Rae (one each). Pia's locker is directly to the right of Oz's (one number higher). Rae's locker is at an end of the row. Quin's locker number is even. Rae's number is lower than Quin's. Oz's locker is not number 2. Who has which locker?

Answer: Oz: 3; Pia: 4; Quin: 2; Rae: 1

Reasoning: Rae is at an end and lower than Quin, so Rae is 1. Oz is not 2, and Pia is directly after Oz, so Oz 3 and Pia 4. Quin gets 2, which is even.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S14-B1-Q1 · question page 310

LOGIC.GRID.01 · Strategy: mark X for "no" and O for "yes" in a grid

Two friends, Ann and Bo, each pick a different color: red or blue. Ann does not pick red. Who picks which color?

Answer: Ann: blue; Bo: red

Reasoning: Ann: blue, so Bo: red.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 2 assignments enumerated; 1 satisfy every clue.

S14-B1-Q2 · question page 311

LOGIC.GRID.01 · Strategy: mark X for "no" and O for "yes" in a grid

In a 3 by 3 grid (3 people, 3 jobs, one each), a person's row has X in two cells. What follows?

Answer: B. The person has the job in the third cell.

Reasoning: Only one option is left in that row.

- ✗ The person has no job.: everyone has exactly one job
- ✗ Nothing follows yet.: only one option remains
- ✗ Another person has that job.: it must be this person's job

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S14-B1-Q3 · question page 312

LOGIC.GRID.01 · Strategy: mark X for "no" and O for "yes" in a grid

In a matching puzzle where each option is used once, if Kim has the bike, no one else has the bike.

Answer: True

Reasoning: Each option is used exactly once.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S14-B1-Q4 · question page 313

LOGIC.GRID.01 · Strategy: mark X for "no" and O for "yes" in a grid

Three cousins, Cy, Di and Ed, are ages 8, 9 and 10 (all different). Cy is the youngest. Ed is not 10. How old is each?

Answer: Cy: 8; Di: 10; Ed: 9

Reasoning: Cy 8; Ed not 10, so 9; Di 10.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S14-B1-Q5 · question page 314

LOGIC.GRID.01 · Strategy: mark X for "no" and O for "yes" in a grid

Three people and three snacks, one each. Gil does not have the apple or the pear. Which snack does Gil have, if the snacks are apple, pear and plum?

Answer: D. plum

Reasoning: Two snacks are ruled out, so Gil has the plum.

✗ apple: ruled out

✗ pear: ruled out

✗ it cannot be known: only plum is left

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S14-F1-Q1 · question page 315

LOGIC.GRID.01 · Grids with relative clues

Four friends, Ari, Bea, Cal and Dev, finished a quiz with scores 6, 7, 8 and 9 (all different). Bea scored 1 more than Dev. Ari scored the most. Cal did not score 6. What did each score?

Answer: Ari: 9; Bea: 7; Cal: 8; Dev: 6

Reasoning: Ari 9. Bea and Dev are consecutive with Bea higher, among 6, 7, 8. Cal is not 6, so Dev is 6 and Bea 7, leaving Cal 8.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S14-F1-Q2 · question page 316

LOGIC.GRID.01 · Grids with relative clues

Five boxes in a row, positions 1 to 5, hold a hat, a kite, a lamp, a mug and a net (one each). The kite is in an even position. The lamp is directly left of the mug. The hat is at position 5. The net is not next to the kite. Which item is in each position?

Answer: 1: net; 2: lamp; 3: mug; 4: kite; 5: hat

Reasoning: The hat is 5th. The kite is 2nd or 4th. Lamp-mug is a pair. If the kite were 2nd, the pair would be 3-4 and the net 1st, next to the kite: not allowed. So the kite is 4th, the pair is 1-2 or 2-3; the net is not next to the kite (not 3rd), so the pair is 2-3 and the net is 1st.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 120 assignments enumerated; 1 satisfy every clue.

S14-F1-Q3 · question page 317

LOGIC.GRID.01 · Grids with relative clues

Rae, Sol and Tom own a cat, a dog and a fish, one each. Clue 1: Rae does not own the cat. Clue 2: Sol does not own the dog. These two clues allow three matchings. Which additional clue leaves exactly one?

Answer: B. Tom owns the cat.

Reasoning: The three matchings allowed by clues 1 and 2 are (Rae, Sol, Tom) = (dog, cat, fish), (dog, fish, cat), (fish, cat, dog). Only "Tom owns the cat" keeps exactly one: (dog, fish, cat).

- ✗ Sol does not own the fish.: two matchings remain: Rae dog, Sol cat, Tom fish; and Rae fish, Sol cat, Tom dog
- ✗ Tom does not own the dog.: two remain: Rae dog with Sol cat or Sol fish
- ✗ Rae does not own the fish.: Rae owns the dog, and Sol and Tom can still split cat and fish two ways

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: each added clue tested against all 6 matchings. Automatic: the check function accepts only the keyed choice.

S14-F1-Q4 · question page 318

LOGIC.GRID.01 · Grids with relative clues

Three students, Uma, Vik and Wyn, each chose a different sport (golf, judo or swimming) and a different day (Monday, Tuesday, Wednesday). The judo student trains on Monday. Uma trains on Wednesday. Uma does not play golf. Wyn trains on Tuesday. Which sport did each choose?

Answer: Uma: swimming; Vik: judo; Wyn: golf

Reasoning: Uma trains Wednesday and Wyn Tuesday, so Vik trains Monday; the judo student trains Monday, so Vik does judo. Uma does not play golf, so Uma swims and Wyn plays golf.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S14-F1-Q5 · question page 319

LOGIC.GRID.01 · Grids with relative clues

In a matching puzzle with 4 people and 4 options (one each), if three people's options are known, the fourth person's option is also known.

Answer: True

Reasoning: Only one option is left.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S14-F2-Q1 · question page 320

LOGIC.GRID.01 · Two-attribute grids and decisive clues

Four students, Al, Bo, Cy and Di, each have a different favorite number from 1 to 4 and a different color (red, blue, green, gold). The red student's number is 1 more than the blue student's number. Al is gold. Bo's number is 4. Cy's number is twice Di's number. What is each student's number?

Answer: Al: 3; Bo: 4; Cy: 2; Di: 1

Reasoning: Bo is 4. Cy = 2 x Di with 1, 2, 3 left gives Di 1 and Cy 2, so Al 3. (The colors are not asked, but they must also work: Al is gold; red is one more than blue, which among Bo 4, Cy 2, Di 1 means Cy red and Di blue, since 4 = 3 + 1 would need Al blue. So Bo is green.)

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S14-F2-Q2 · question page 321

LOGIC.GRID.01 · Two-attribute grids and decisive clues

Three people wear a red, a blue and a green hat, one each, so there are 6 possible matchings. Clues have the form "person X does not wear color Y". What is the smallest number of such clues that can leave exactly one matching?

Answer: A. 3

Reasoning: Example with 3 clues: A not red, A not blue (so A green), B not red (so B blue): C red. With 2 clues, at least 2 matchings always remain (checked for every pair of clues).

- ✗ 1: one clue removes 2 matchings and leaves 4
- ✗ 2: two clues always leave at least 2 matchings
- ✗ 4: three well-chosen clues already suffice

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: every set of 1, 2 and 3 clues tested against the 6 matchings. Automatic: the check function accepts only the keyed choice.

S14-F2-Q3 · question page 322

LOGIC.GRID.01 · Two-attribute grids and decisive clues

Four people each get a different one of 4 prizes. How many matchings are possible if one person, Ash, cannot get prize 1 or prize 2?

Answer: 12

Reasoning: Ash has 2 choices (prize 3 or 4); the other three people take the remaining 3 prizes in 3 x 2 x 1 = 6 ways: 2 x 6 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S14-F2-Q4 · question page 323

LOGIC.GRID.01 · Two-attribute grids and decisive clues

In a 3-person, 3-option matching puzzle, the clues "A not x" and "B not x" together force C to

have x.

Answer: True

Reasoning: Option x must go to someone, and A and B are ruled out.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S14-F2-Q5 · question page 324

LOGIC.GRID.01 · Two-attribute grids and decisive clues

Five players wear the numbers 1 to 5 (all different). Al's number plus Bo's number is 9. Give one possible pair: Al's number a and Bo's number b.

Answer: any values that satisfy: a + b == 9 and a != b and a >= 1 and a <= 5 and b >= 1 and b <= 5 and isInteger(a) and isInteger(b) (example: a = 4, b = 5)

Reasoning: Only 4 + 5 = 9 works with different numbers up to 5, in either order.

Tempting answers the rule rejects: a = 3, b = 6; a = 5, b = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 2 valid answers.

S15-M-Q1 · question page 326

LOGIC.CONDITIONAL.03 · Form the converse, inverse and contrapositive and judge equivalence

What is the CONTRAPOSITIVE of "If a shape is a triangle, then it has three sides"?

Answer: C. If a shape does not have three sides, then it is not a triangle.

Reasoning: The contrapositive of "if P then Q" is "if not Q, then not P".

✗ If a shape has three sides, then it is a triangle.: that is the converse

✗ If a shape is not a triangle, then it does not have three sides.: that is the inverse

✗ A shape is a triangle and does not have three sides.: that is the negation of the conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-M-Q2 · question page 327

LOGIC.CONDITIONAL.03 · Form the converse, inverse and contrapositive and judge equivalence

Which statement is logically EQUIVALENT to "If n is a multiple of 4, then n is even"?

Answer: D. If n is not even, then n is not a multiple of 4.

Reasoning: A conditional and its contrapositive always have the same truth value.

✗ If n is even, then n is a multiple of 4.: the converse is not equivalent (6 is even)

✗ If n is not a multiple of 4, then n is not even.: the inverse is not equivalent (6 is even)

✗ n is a multiple of 4 and n is even.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S15-M-Q3 · question page 328

LOGIC.CONDITIONAL.03 · Form the converse, inverse and contrapositive and judge equivalence

The converse of "If a whole number is a multiple of 6, then it is even" is true.

Answer: False

Reasoning: The converse, "If a whole number is even, then it is a multiple of 6", is false: 4 is even but not a multiple of 6.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S15-M-Q4 · question page 329

LOGIC.CONDITIONAL.03 · Form the converse, inverse and contrapositive and judge equivalence

The statement "If x > 2, then x squared > 4" is true. Find a real number x that shows its CONVERSE is false.

Answer: any values that satisfy: x^2 > 4 and not (x > 2) (example: x = -3)

Reasoning: The converse is "If x squared > 4, then x > 2". It fails for any x < -2, for example -3.

Tempting answers the rule rejects: x = 3; x = 1; x = -2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S15-M-Q5 · question page 330

LOGIC.CONDITIONAL.03 · Form the converse, inverse and contrapositive and judge equivalence

"If P, then Q" is true. Which statement MUST also be true?

Answer: A. If not Q, then not P.

Reasoning: Only the contrapositive is equivalent to the original.

- ✗ If Q, then P.: the converse can be false
- ✗ If not P, then not Q.: the inverse can be false

✗ P and Q.: P may be false

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: checked in every row where "if P then Q" is true. Automatic: the check function accepts only the keyed choice.

S15-B1-Q1 · question page 331

LOGIC.CONNECTIVE.01 · Negate each part

What is the negation of "the shape has three sides"?

Answer: C. the shape does not have three sides

Reasoning: Put "not" on the whole claim.

✗ the shape has four sides: too specific
✗ the shape has no sides: too specific
✗ the shape has at least three sides: not a negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-B1-Q2 · question page 332

LOGIC.CONNECTIVE.01 · Negate each part

What is the negation of "n is even" for a whole number n?

Answer: B. n is odd

Reasoning: Whole numbers that are not even are odd.

✗ n is zero: too specific
✗ n is prime: unrelated
✗ n is negative: unrelated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S15-B1-Q3 · question page 333

LOGIC.CONNECTIVE.01 · Negate each part

The negation of "x > 3" is "x < 3".

Answer: False

Reasoning: It is "x ≤ 3".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S15-B1-Q4 · question page 334

LOGIC.CONNECTIVE.01 · Negate each part

What is the negation of "it is raining"?

Answer: A. it is not raining

Reasoning: Negate the whole claim.

- ✗ it is sunny: it could be cloudy and dry
- ✗ it rained yesterday: unrelated
- ✗ it will rain: unrelated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-B1-Q5 · question page 335

LOGIC.CONNECTIVE.01 · Negate each part

"Not (not P)" says the same as "not P".

Answer: False

Reasoning: Two negations cancel, so "not (not P)" says the same as P.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S15-B2-Q1 · question page 336

LOGIC.CONDITIONAL.01 · Hypothesis and conclusion

In "If it snows, school is closed," what is the conclusion?

Answer: B. school is closed

Reasoning: The "then" part.

- ✗ it snows: the hypothesis
- ✗ it is cold: not stated
- ✗ school is open: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-B2-Q2 · question page 337

LOGIC.CONDITIONAL.01 · Hypothesis and conclusion

Swap the parts of "If P, then Q". What do you get?

Answer: D. If Q, then P.

Reasoning: Swapping gives the converse.

- ✗ If not P, then not Q.: that negates instead of swapping
- ✗ If not Q, then not P.: that swaps AND negates
- ✗ P and Q.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-B2-Q3 · question page 338

LOGIC.CONDITIONAL.01 · Hypothesis and conclusion

In "If x = 5, then x + 1 = 6," the conclusion is "x = 5".

Answer: False

Reasoning: The conclusion is the part after "then": "x + 1 = 6". "x = 5" is the hypothesis.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S15-B2-Q4 · question page 339

LOGIC.CONDITIONAL.01 · Hypothesis and conclusion

Which statement has hypothesis "a figure is a square"?

Answer: C. If a figure is a square, then it is a rectangle.

Reasoning: The hypothesis is the "if" part: "a figure is a square".

- ✗ If a figure is a rectangle, then it is a square.: hypothesis is "rectangle"
- ✗ A figure is a rectangle only if it has four sides.: hypothesis is "a figure is a rectangle"
- ✗ Every rectangle is a square.: hypothesis is "rectangle"

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-B2-Q5 · question page 340

LOGIC.CONDITIONAL.01 · Hypothesis and conclusion

Swapping and negating both parts of "If P, then Q" gives "If not Q, then not P".

Answer: True

Reasoning: That is the contrapositive.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S15-F1-Q1 · question page 341

LOGIC.CONDITIONAL.03 · Which forms are equivalent?

Complete the truth table for "if P then Q" and its contrapositive "if not Q then not P".

Answer: T T / F F / T T / T T

Reasoning: The two columns are identical: a conditional and its contrapositive are equivalent.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S15-F1-Q2 · question page 342

LOGIC.CONDITIONAL.03 · Which forms are equivalent?

Which two statements are ALWAYS equivalent to each other?

Answer: A. the converse and the inverse

Reasoning: The inverse "if not P then not Q" is the contrapositive of the converse "if Q then P".

- ✗ the original and the converse: they differ when P is false and Q true
- ✗ the original and the inverse: they differ when P is false and Q true
- ✗ the converse and the contrapositive: they differ when P is false and Q true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S15-F1-Q3 · question page 343

LOGIC.CONDITIONAL.03 · Which forms are equivalent?

For how many whole numbers n from 1 to 20 is the INVERSE of "If n is a multiple of 4, then n is even" false?

Answer: 5

Reasoning: The inverse is "If n is not a multiple of 4, then n is not even". It is false when n is even but not a multiple of 4: 2, 6, 10, 14, 18.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S15-F1-Q4 · question page 344

LOGIC.CONDITIONAL.03 · Which forms are equivalent?

If a conditional is false, its contrapositive is true.

Answer: False

Reasoning: A conditional and its contrapositive are equivalent, so they are always true or false together.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S15-F1-Q5 · question page 345

LOGIC.CONDITIONAL.03 · Which forms are equivalent?

"If a quadrilateral is a square, then its diagonals are equal." Which related statement is FALSE?

Answer: B. the converse: if the diagonals are equal, the quadrilateral is a square

Reasoning: A rectangle that is not a square has equal diagonals, so the converse is false.

✗ the contrapositive: if the diagonals are not equal, it is not a square: equivalent to the true original

✗ the original statement: true for every square

✗ "a square has equal diagonals": the original, restated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-F2-Q1 · question page 346

LOGIC.CONDITIONAL.03 · Converses that hold and fail

For whole numbers, both "If n is odd, then n squared is odd" and its converse are true.

Answer: True

Reasoning: An odd number squared is odd; and if n were even, n squared would be even, so n squared odd forces n odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S15-F2-Q2 · question page 347

LOGIC.CONDITIONAL.03 · Converses that hold and fail

The statement "If n is prime and greater than 2, then n is odd" is true. Its converse is "If n is odd, then n is prime and greater than 2". Give a whole number n that shows the converse is false.

Answer: any values that satisfy: isOdd(n) and not (isPrime(n) and n > 2) (example: n = 9)

Reasoning: Any odd number that is not a prime greater than 2: 1, 9, 15, 21, ...

Tempting answers the rule rejects: n = 7; n = 4; n = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are

correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S15-F2-Q3 · question page 348

LOGIC.CONDITIONAL.03 · Converses that hold and fail

A statement and its converse are both true. Which statement must be true?

Answer: C. P if and only if Q

Reasoning: If P -> Q and Q -> P, then P and Q always match: that is "P if and only if Q".

- ✗ P and Q: both could be false
- ✗ not P: P could be true
- ✗ P or Q: both could be false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S15-F2-Q4 · question page 349

LOGIC.CONDITIONAL.03 · Converses that hold and fail

In how many of the 4 rows of a truth table do "if P then Q" and its converse "if Q then P" have DIFFERENT truth values?

Answer: 2

Reasoning: They differ when exactly one of P, Q is true: rows TF and FT.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S15-F2-Q5 · question page 350

LOGIC.CONDITIONAL.03 · Converses that hold and fail

C is the conditional "if P then Q". Which statement is correct?

Answer: D. If C is true, its inverse may be true or false.

Reasoning: In the rows where C is true, the inverse is true for (T, T) and (F, F) and false for (F, T), so it may be either.

- ✗ If C is true, its converse is true.: when P is false and Q true, C is true but the converse is false
- ✗ If C is false, its converse is false.: C is false only when P is true and Q false; then the converse "if Q then P" is true
- ✗ If C is true, its contrapositive may be false.: never: they are equivalent

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: each claim tested in all four rows. Automatic: the check function accepts only the keyed choice.

S16-M-Q1 · question page 352

LOGIC.CONDITIONAL.04 · Find truth values of "if and only if" statements and use them as definitions

Complete the truth table for "P if and only if Q".

Answer: T / F / F / T

Reasoning: It is true exactly when P and Q have the same truth value.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S16-M-Q2 · question page 353

LOGIC.CONDITIONAL.04 · Find truth values of "if and only if" statements and use them as definitions

For real numbers x, "x squared = 9 if and only if x = 3" is true.

Answer: False

Reasoning: x = -3 makes the left part true and the right part false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-M-Q3 · question page 354

LOGIC.CONDITIONAL.04 · Find truth values of "if and only if" statements and use them as definitions

Find a real number x that makes "x > 5 if and only if x > 3" FALSE.

Answer: any values that satisfy: (x > 5) != (x > 3) (example: x = 4)

Reasoning: The two parts must differ: x > 3 but not x > 5, so any x with 3 < x ≤ 5.

Tempting answers the rule rejects: x = 6; x = 2; x = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S16-M-Q4 · question page 355

LOGIC.CONDITIONAL.04 · Find truth values of "if and only if" statements and use them as definitions

Which statement is a correct DEFINITION of an even whole number?

Answer: B. A whole number is even if and only if it is divisible by 2.

Reasoning: A definition must work both ways: every even number is divisible by 2, and every number divisible by 2 is even.

✗ A whole number is even if it is divisible by 4.: true, but it leaves out 2, 6, 10, ...

✗ A whole number is even only if it is greater than 1.: 0 is even

✗ A whole number is even if and only if it ends in 0.: 4 is even and does not end in 0

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-M-Q5 · question page 356

LOGIC.CONDITIONAL.04 · Find truth values of "if and only if" statements and use them as definitions

"P if and only if Q" is true and P is false. What follows?

Answer: C. Q is false.

Reasoning: In a true biconditional the two parts have the same value.

✗ Q is true.: then P and Q would differ

✗ Nothing about Q.: Q must match P

✗ P and Q are both true.: P is false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-B1-Q1 · question page 357

LOGIC.CONDITIONAL.01 · Conditionals first

"If 2 > 3, then 4 > 5" is true.

Answer: True

Reasoning: The "if" part is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-B1-Q2 · question page 358

LOGIC.CONDITIONAL.01 · Conditionals first

"If 3 > 2, then 4 > 5" is true.

Answer: False

Reasoning: True "if", false "then".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-B1-Q3 · question page 359

LOGIC.CONDITIONAL.01 · Conditionals first

When is "if P then Q" false?

Answer: A. when P is true and Q is false

Reasoning: Only one row is false.

- ✗ when P is false: it is true then
- ✗ when Q is true: it is true then
- ✗ when both are false: it is true then

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-B1-Q4 · question page 360

LOGIC.CONDITIONAL.01 · Conditionals first

"If 4 > 1, then 2 > 6" is true.

Answer: False

Reasoning: The "if" part is true and the "then" part is false, which is the

one case that makes a conditional false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-B1-Q5 · question page 361

LOGIC.CONDITIONAL.01 · Conditionals first

In how many of the 4 rows is "if P then Q" true?

Answer: 3

Reasoning: All rows except P true, Q false.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S16-B2-Q1 · question page 362

LOGIC.CONDITIONAL.03 · The converse

What is the converse of "If it is a dog, then it barks"?

Answer: D. If it barks, then it is a dog.

Reasoning: Swap the parts.

- ✗ If it is not a dog, then it does not bark.: the inverse
- ✗ If it does not bark, then it is not a dog.: the contrapositive
- ✗ It is a barking dog.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-B2-Q2 · question page 363

LOGIC.CONDITIONAL.03 · The converse

The converse of "If n = 2, then n is even" is "If n is not 2, then n is not even".

Answer: False

Reasoning: That statement negates both parts; it is the inverse. The converse swaps the parts: "If n is even, then n = 2".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-B2-Q3 · question page 364

LOGIC.CONDITIONAL.03 · The converse

The converse of a true statement is always true.

Answer: False

Reasoning: Example: "If n = 2, then n is even" is true, its converse is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-B2-Q4 · question page 365

LOGIC.CONDITIONAL.03 · The converse

"P if and only if Q" means "if P then Q" AND what?

Answer: A. if Q then P

Reasoning: The other direction is the converse.

- ✗ if not P then Q: not the converse
- ✗ P or Q: not part of it
- ✗ not P and not Q: not part of it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-B2-Q5 · question page 366

LOGIC.CONDITIONAL.03 · The converse

If a conditional is true but its converse is false, the biconditional is true.

Answer: False

Reasoning: A biconditional needs both directions true; a false converse makes it false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-F1-Q1 · question page 367

LOGIC.CONDITIONAL.04 · Biconditionals as definitions and tests

For how many whole numbers n from 1 to 20 is "n is even if and only if n is a multiple of 4" TRUE?

Answer: 15

Reasoning: It fails exactly when n is even but not a multiple of 4: 2, 6, 10, 14, 18 (5 numbers). The other 15 make it true (odd numbers make both parts false).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S16-F1-Q2 · question page 368

LOGIC.CONDITIONAL.04 · Biconditionals as definitions and tests

Complete the table to compare "P if and only if Q" with "(if P then Q) and (if Q then P)".

Answer: T T / F F / F F / T T

Reasoning: The columns match in every row.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S16-F1-Q3 · question page 369

LOGIC.CONDITIONAL.04 · Biconditionals as definitions and tests

Which definition of a SQUARE is correct (works in both directions)?

Answer: B. A quadrilateral is a square if and only if it has four equal sides and four right angles.

Reasoning: Only the first condition is true of every square and of nothing else.

- ✗ A quadrilateral is a square if and only if it has four equal sides.: a rhombus with no right angles has four equal sides
- ✗ A quadrilateral is a square if and only if it has four right angles.: a non-square rectangle has four right angles
- ✗ A quadrilateral is a square if and only if its diagonals are equal.: a non-square rectangle has equal diagonals

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-F1-Q4 · question page 370

LOGIC.CONDITIONAL.04 · Biconditionals as definitions and tests

"A whole number is divisible by 10 if and only if it ends in 0" is true.

Answer: True

Reasoning: Both directions hold for whole numbers.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S16-F1-Q5 · question page 371

LOGIC.CONDITIONAL.04 · Biconditionals as definitions and tests

Find a whole number n that makes "n is divisible by 3 if and only if n is odd" FALSE.

Answer: any values that satisfy: (n % 3 == 0) != isOdd(n) and isInteger(n) (example: n = 6)

Reasoning: The parts must differ: divisible by 3 but even (6, 12, ...) or odd but not divisible by 3 (1, 5, ...).

Tempting answers the rule rejects: n = 9; n = 4; n = 15

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S16-F2-Q1 · question page 372

LOGIC.CONDITIONAL.04 · Chains and opposites of biconditionals

"P exclusive-or Q" has the opposite truth value of "P if and only if Q" in every row.

Answer: True

Reasoning: Exclusive-or is true exactly when the parts differ; the biconditional exactly when they match.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S16-F2-Q2 · question page 373

LOGIC.CONDITIONAL.04 · Chains and opposites of biconditionals

"P iff Q" and "Q iff R" are both true. Which must be true?

Answer: C. P iff R

Reasoning: P matches Q and Q matches R, so P matches R.

- ✗ P and R: all three could be false
- ✗ P exclusive-or R: P and R match
- ✗ not Q: Q could be true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S16-F2-Q3 · question page 374

LOGIC.CONDITIONAL.04 · Chains and opposites of biconditionals

In how many of the 8 rows of a truth table for P, Q, R is "(P iff Q) iff R" true?

Answer: 4

Reasoning: Exactly half of the rows: the statement is true when an even number of P, Q, R are false.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S16-F2-Q4 · question page 375

LOGIC.CONDITIONAL.04 · Chains and opposites of biconditionals

Find whole numbers a and b, each from 1 to 9, that make "a + b is even if and only if a is even" TRUE and a different from b.

Answer: any values that satisfy: (isEven(a + b) == isEven(a)) and a != b and a >= 1 and a <= 9 and b >= 1 and b <= 9 (example: a = 2, b = 4)

Reasoning: The biconditional is true exactly when b is even (then a + b and a have the same parity). Any a, b with b even and a different from b

works, such as 2 and 4 or 3 and 8.

Tempting answers the rule rejects: a = 2, b = 3; a = 4, b = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 32 valid answers.

S16-F2-Q5 · question page 376

LOGIC.CONDITIONAL.04 · Chains and opposites of biconditionals

Which statement about whole numbers is TRUE?

Answer: D. n is odd if and only if n + 1 is even.

Reasoning: Adding 1 always changes parity, so n odd and n + 1 even always match.

- ✗ n is odd if and only if n is prime.: 9 is odd, not prime; 2 is prime, not odd
- ✗ n is even if and only if n is a multiple of 4.: 6 is even but not a multiple of 4
- ✗ n is a multiple of 3 if and only if n is a multiple of 9.: 3 is a multiple of 3 but not of 9

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S17-M-Q1 · question page 378

LOGIC.KNIGHTS.01 · Decide who tells the truth when each speaker always tells the truth or always lies

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Ana says: "Ben is a liar." Ben says: "Ana and I are both truth-tellers." Mark each person as a truth-teller or a liar.

Answer: Ana: truth-teller; Ben: liar

Reasoning: If Ben were truthful, both would be truthful, but then Ana's "Ben is a liar" would be false. So Ben lies, Ana's statement is true, and Ana is a truth-teller.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 4 assignments enumerated; 1 satisfy every clue.

S17-M-Q2 · question page 379

LOGIC.KNIGHTS.01 · Decide who tells the truth when each speaker always tells the truth or always lies

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Cal says: "Dee and I are different kinds." Dee says: "Cal is a liar." Mark each person as a truth-teller or a liar.

Answer: Cal: truth-teller; Dee: liar

Reasoning: Suppose Cal tells the truth: they are different, so Dee is a liar, and Dee's claim "Cal is a liar" is indeed false. Consistent. Suppose Cal lies: they are the same kind, so Dee lies too, but then "Cal is a liar" would be true, a contradiction. So Cal is truthful and Dee lies.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 4 assignments enumerated; 1 satisfy every clue.

S17-M-Q3 · question page 380

LOGIC.KNIGHTS.01 · Decide who tells the truth when each speaker always tells the truth or always lies

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Eve says: "Fin and Gia are both liars." Fin says: "Eve is a truth-teller." Gia says: "Fin is a liar." Mark each person as a truth-teller or a liar.

Answer: Eve: liar; Fin: liar; Gia: truth-teller

Reasoning: Fin and Eve stand or fall together (Fin says Eve is truthful). If both were truthful, Eve's claim would make Fin a liar: contradiction. So Eve and Fin are liars. Then Gia's "Fin is a liar" is true, so Gia is truthful (and Eve's claim is indeed false).

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 8 assignments enumerated; 1 satisfy every clue.

S17-M-Q4 · question page 381

LOGIC.KNIGHTS.01 · Decide who tells the truth when each speaker always tells the truth or always lies

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Jo says: "Kai is a liar." Kai says: "Lu is a liar." Lu says: "Jo and Kai are both liars." Mark each person as a truth-teller or a liar.

Answer: Jo: liar; Kai: truth-teller; Lu: liar

Reasoning: If Lu were truthful, Jo and Kai would both lie, but Jo lying means Kai is truthful: contradiction. So Lu lies; then Kai's "Lu is a liar" is true, Kai is truthful, and Jo (who calls Kai a liar) lies.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 8 assignments enumerated; 1 satisfy every clue.

S17-M-Q5 · question page 382

LOGIC.KNIGHTS.01 · Decide who tells the truth when each speaker always tells the truth or always lies

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Mo says: "Exactly one of Ned and Ola is a liar." Ned says: "Mo is a liar." Ola says: "Ned and I are the same kind." Who is a liar?

Answer: B. Mo only

Reasoning: Check the 8 possibilities: only Mo lying, with Ned and Ola truthful, is consistent. Ned truthfully says Mo lies; Ola truthfully says Ned

and Ola match; Mo's claim "exactly one of Ned and Ola lies" is false.

✗ Ned only: then Mo and Ola are truthful, but Ola would be claiming Ned and Ola are the same kind

✗ Ola only: then Ned is truthful and calls Mo a liar, but Mo would be truthful

✗ Mo and Ned: then Ola is truthful, but Ola says Ned and Ola are the same kind

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 8 assignments checked. Automatic: the check function accepts only the keyed choice.

S17-B1-Q1 · question page 383

LOGIC.KNIGHTS.01 · Strategy: suppose, then test

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A truth-teller can say "I am a liar."

Answer: False

Reasoning: That would be a false statement.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S17-B1-Q2 · question page 384

LOGIC.KNIGHTS.01 · Strategy: suppose, then test

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A liar can say "I am a truth-teller."

Answer: True

Reasoning: That is a false statement, which a liar can make.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S17-B1-Q3 · question page 385

LOGIC.KNIGHTS.01 · Strategy: suppose, then test

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Vic says: "2 + 2 = 5." What is Vic?

Answer: C. a liar

Reasoning: A false statement comes from a liar.

- ✗ a truth-teller: the statement is false
- ✗ either: only liars make false statements
- ✗ neither: everyone is one or the other

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S17-B1-Q4 · question page 386

LOGIC.KNIGHTS.01 · Strategy: suppose, then test

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Wes is a truth-teller and says: "Xia is a liar." What is Xia?

Answer: D. a liar

Reasoning: Wes tells the truth.

- ✗ a truth-teller: Wes speaks truly
- ✗ either: Wes's true statement decides it
- ✗ neither: everyone is one or the other

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S17-B1-Q5 · question page 387

LOGIC.KNIGHTS.01 · Strategy: suppose, then test

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Yan is a liar and says: "Zed is a liar." What is Zed?

Answer: A. a truth-teller

Reasoning: The statement is false, so Zed is not a liar.

- ✗ a liar: Yan's statement is false
- ✗ either: it is decided
- ✗ neither: everyone is one or the other

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S17-F1-Q1 · question page 388

LOGIC.KNIGHTS.01 · Statements about counts

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Ash says: "Exactly one of us three is a truth-teller." Bly says: "Exactly two of us three are truth-tellers." Cru says: "None of us three is a truth-teller." Mark each person as a truth-teller or a liar.

Answer: Ash: truth-teller; Bly: liar; Cru: liar

Reasoning: The three claims contradict each other, so at most one is true. Cru cannot be truthful (that would make none truthful). If none were truthful, Cru's claim would be true: contradiction. So exactly one is truthful, which makes Ash's claim true: Ash is the truth-teller.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 8 assignments enumerated; 1 satisfy every clue.

S17-F1-Q2 · question page 389

LOGIC.KNIGHTS.01 · Statements about counts

On this island every person is either a truth-teller, who always tells the truth, or a liar, who

always lies. Dov says: "At least one of Ela and Fyn is a liar." Ela says: "Dov and Fyn are both truth-tellers." Fyn says: "Dov is a liar." Mark each person as a truth-teller or a liar.

Answer: Dov: truth-teller; Ela: liar; Fyn: liar

Reasoning: Suppose Dov truthful: then Fyn ("Dov is a liar") lies, so Ela's claim is false and Ela lies; Dov's claim (one of Ela, Fyn lies) is true. Consistent. Suppose Dov lies: then Ela and Fyn are both truthful, but Ela says Dov is truthful: contradiction. So Dov is truthful, Ela and Fyn lie.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 8 assignments enumerated; 1 satisfy every clue.

S17-F1-Q3 · question page 390

LOGIC.KNIGHTS.01 · Statements about counts

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Pat says: "Exactly two of us three are truth-tellers." Quy says: "Pat is a liar." Ren says: "Quy and Pat are the same kind." Mark each person as a truth-teller or a liar.

Answer: Pat: liar; Quy: truth-teller; Ren: liar

Reasoning: Quy and Pat are always different kinds (Quy calls Pat a liar), so Ren's claim is false: Ren lies. Then exactly one of Pat, Quy is truthful, so the number of truth-tellers is 1, Pat's claim is false, Pat lies and Quy is truthful.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 8 assignments enumerated; 1 satisfy every clue.

S17-F1-Q4 · question page 391

LOGIC.KNIGHTS.01 · Statements about counts

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. If two people each say "The other one is a liar," then exactly one of them is a liar.

Answer: True

Reasoning: If A is truthful, B lies; if A lies, B is truthful. Either way they are different.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S17-F1-Q5 · question page 392

LOGIC.KNIGHTS.01 · Statements about counts

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Four people each say exactly the same sentence: "Exactly one of us four is a truth-teller." How many of them are truth-tellers?

Answer: 0

Reasoning: They say the same sentence, so they are all the same kind. If all were truthful, there would be four truth-tellers, not one: impossible. So all four lie, and indeed the sentence is false (there are none).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S17-F2-Q1 · question page 393

LOGIC.KNIGHTS.01 · Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Uri says: "Exactly three of us four are liars." Vee says: "Uri is a truth-teller." Wil says: "Vee is a liar." Xan says: "Wil is a truth-teller and Uri is a liar." Mark each person as a truth-teller or a liar.

Answer: Uri: liar; Vee: liar; Wil: truth-teller; Xan: truth-teller

Reasoning: Vee and Uri are the same kind; Wil is the opposite of Vee. If Uri were truthful, Vee would be too, and at most two could lie:

contradiction with "exactly three". So Uri and Vee lie and Wil is truthful. Xan's claim (Wil truthful, Uri lying) is true, so Xan is truthful. Two liars: Uri's claim is false, as it must be.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 16 assignments enumerated; 1 satisfy every clue.

S17-F2-Q2 · question page 394

LOGIC.KNIGHTS.01 · Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Which statement can NO person on the island say?

Answer: A. "I am a liar."

Reasoning: A truth-teller saying it would be lying; a liar saying it would be telling the truth.

- ✗ "I am a truth-teller.": anyone can say it: true for a truth-teller, false for a liar
- ✗ "2 + 2 = 4.": a truth-teller can say it
- ✗ "2 + 2 = 5.": a liar can say it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S17-F2-Q3 · question page 395

LOGIC.KNIGHTS.01 · Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Five people stand in a circle. Each says: "The person on my left is a liar." In how many ways can the five be truth-tellers and liars so that every statement fits?

Answer: 0

Reasoning: Each statement makes neighbors different kinds, so the types must alternate around the circle. An odd circle cannot alternate

(going once around returns to the start after 5 changes), so no assignment works.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S17-F2-Q4 · question page 396

LOGIC.KNIGHTS.01 · Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A person says: "I am a liar or 2 + 2 = 4." This person must be a truth-teller.

Answer: True

Reasoning: The statement is true (2 + 2 = 4), so the speaker tells the truth. A liar could not say a true statement.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S17-F2-Q5 · question page 397

LOGIC.KNIGHTS.01 · Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. You meet Gil and Hal. Gil says: "At least one of us two is a truth-teller." Hal says nothing. Which statement MUST be true?

Answer: D. If Hal is a truth-teller, then Gil is a truth-teller.

Reasoning: Three cases fit: both truthful; Gil truthful and Hal lying; both lying. In every case where Hal is truthful, Gil's claim is true, so Gil is truthful.

- ✗ Gil is a truth-teller.: Gil and Hal could both be liars: then Gil's claim is false, as a liar's must be
- ✗ Hal is a liar.: Gil and Hal could both be truth-tellers

✗ Both are liars.: Gil could be a truth-teller

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 4 assignments checked; 3 fit. Automatic: the check function accepts only the keyed choice.

S18-M-Q1 · question page 399

LOGIC.TRUTH.01 · Complete truth tables for compound statements

Complete the truth table for "P and (not Q)".

Answer: F F / T T / F F / T F

Reasoning: First fill "not Q", then combine with P: true only when P is true and Q is false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-M-Q2 · question page 400

LOGIC.TRUTH.01 · Complete truth tables for compound statements

Complete the truth table for "(not P) or Q".

Answer: F T / F F / T T / T T

Reasoning: False only when P is true and Q is false (it matches "if P then Q").

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-M-Q3 · question page 401

LOGIC.TRUTH.01 · Complete truth tables for compound statements

Complete the truth table for "(P or Q) and not (P and Q)".

Answer: T T F / T F T / T F T / F F F

Reasoning: True exactly when one of P, Q is true: exclusive or.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-M-Q4 · question page 402

LOGIC.TRUTH.01 · Complete truth tables for compound statements

A truth table for three statements P, Q and R lists every combination of truth values. How many rows does it have?

Answer: 8

Reasoning: Each statement has 2 values: 2 x 2 x 2 = 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S18-M-Q5 · question page 403

LOGIC.TRUTH.01 · Complete truth tables for compound statements

Complete the truth table for "(P and Q) or R".

Answer: T T / T T / F T / F F / F T / F F / F T / F F

Reasoning: True when R is true or both P and Q are true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (8 rows).

S18-B1-Q1 · question page 404

LOGIC.CONNECTIVE.02 · The basic connectives

Complete the truth table for "P and Q".

Answer: T / F / F / F

Reasoning: True only when both are true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-B1-Q2 · question page 405

LOGIC.CONNECTIVE.02 · The basic connectives

Complete the truth table for "P or Q" (inclusive).

Answer: T / T / T / F

Reasoning: False only when both are false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-B1-Q3 · question page 406

LOGIC.CONNECTIVE.02 · The basic connectives

Complete the truth table for "not P".

Answer: F / T

Reasoning: Flip the value.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (2 rows).

S18-B1-Q4 · question page 407

LOGIC.CONNECTIVE.02 · The basic connectives

"P and Q" is true when P is true and Q is false.

Answer: False

Reasoning: Both must be true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S18-B1-Q5 · question page 408

LOGIC.CONNECTIVE.02 · The basic connectives

"P or Q" is true when P is false and Q is true.

Answer: True

Reasoning: At least one is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S18-B2-Q1 · question page 409

LOGIC.TRUTH.01 · Work from the inside out

In "not (P and Q)", which part is computed first?

Answer: C. P and Q

Reasoning: Inside the brackets first.

- ✗ not P: "not" applies to the whole bracket
- ✗ not Q: same reason
- ✗ P or Q: not in the statement

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S18-B2-Q2 · question page 410

LOGIC.TRUTH.01 · Work from the inside out

"not (P and Q)" and "(not P) and Q" always have the same truth value.

Answer: False

Reasoning: P true, Q true: the first is false and the second false; P false, Q false: the first is true and the second false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S18-B2-Q3 · question page 411

LOGIC.TRUTH.01 · Work from the inside out

P is true and Q is false. Is "not (P and Q)" true or false? Enter 1 for true or 0 for false.

Answer: 1

Reasoning: P and Q is false, so its negation is true.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S18-B2-Q4 · question page 412

LOGIC.TRUTH.01 · Work from the inside out

P is false, Q is true. What is the value of "(P or Q) and P"?

Answer: B. false

Reasoning: P or Q is true; true and false is false.

- ✗ true: the final "and P" is false
- ✗ it cannot be found: all parts are known
- ✗ both: a statement has one value

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S18-B2-Q5 · question page 413

LOGIC.TRUTH.01 · Work from the inside out

Complete the truth table for "not (P or Q)".

Answer: T F / T F / T F / F T

Reasoning: Compute the bracket, then negate.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-F1-Q1 · question page 414

LOGIC.TRUTH.01 · Three-variable tables

Complete the truth table for "P and (Q or R)".

Answer: T T / T T / T T / F F / T F / T F / T F / F F

Reasoning: True when P is true and at least one of Q, R is true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (8 rows).

S18-F1-Q2 · question page 415

LOGIC.TRUTH.01 · Three-variable tables

Complete the truth table for "(P or Q) and (not R)".

Answer: T F / T T / T F / T T / T F / T T / F F / F F

Reasoning: True when R is false and at least one of P, Q is true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (8 rows).

S18-F1-Q3 · question page 416

LOGIC.TRUTH.01 · Three-variable tables

In how many of the 8 rows is "P or (Q and R)" true?

Answer: 5

Reasoning: True whenever P is true (4 rows) plus the row P false, Q true, R true: 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S18-F1-Q4 · question page 417

LOGIC.TRUTH.01 · Three-variable tables

A statement in P and Q is true only in the row P true, Q false. Which statement is it?

Answer: A. P and (not Q)

Reasoning: Check each candidate in the four rows.

- ✗ P or (not Q): also true when both are true and when both are false
- ✗ (not P) and Q: true only when P is false and Q true
- ✗ not (P and Q): true in three rows

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S18-F1-Q5 · question page 418

LOGIC.TRUTH.01 · Three-variable tables

A truth table for 4 statements has 16 rows.

Answer: True

Reasoning: 2 x 2 x 2 x 2 = 16.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S18-F2-Q1 · question page 419

LOGIC.TRUTH.01 · Read and build statements from tables

A statement in P, Q, R is true in exactly the rows where an odd number of P, Q, R are true. Which statement is it?

Answer: C. P exclusive-or (Q exclusive-or R)

Reasoning: Exclusive-or flips the value each time a true part is added.

- ✗ P or Q or R: true also when two are true
- ✗ P and Q and R: true only when all three are true
- ✗ (P and Q) or R: true in rows with two true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S18-F2-Q2 · question page 420

LOGIC.TRUTH.01 · Read and build statements from tables

How many different possible output columns can a truth table with two statement letters P and Q have (4 rows, each output T or F)?

Answer: 16

Reasoning: Each of the 4 rows can be T or F: 2 to the 4th = 16.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S18-F2-Q3 · question page 421

LOGIC.TRUTH.01 · Read and build statements from tables

Complete the truth table for "(P and not Q) or (not P and Q)".

Answer: F F F / T F T / F T T / F F F

Reasoning: Each row where exactly one is true makes one bracket true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-F2-Q4 · question page 422

LOGIC.TRUTH.01 · Read and build statements from tables

In how many of the 16 rows of a table for P, Q, R, S is "(P and Q) or (R and S)" true?

Answer: 7

Reasoning: P and Q true: 4 rows; R and S true: 4 rows; both: 1 row. 4 + 4 - 1 = 7.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S18-F2-Q5 · question page 423

LOGIC.TRUTH.01 · Read and build statements from tables

Every truth-table column for P and Q can be produced by some statement using only "not", "and", "or".

Answer: True

Reasoning: For each true row write a bracket like "P and not Q", then join the brackets with "or" (for the all-F column use "P and not P").